

Chart

Chart [object]

Use the object *Chart* for presenting current data and results of the simulation run.

Image

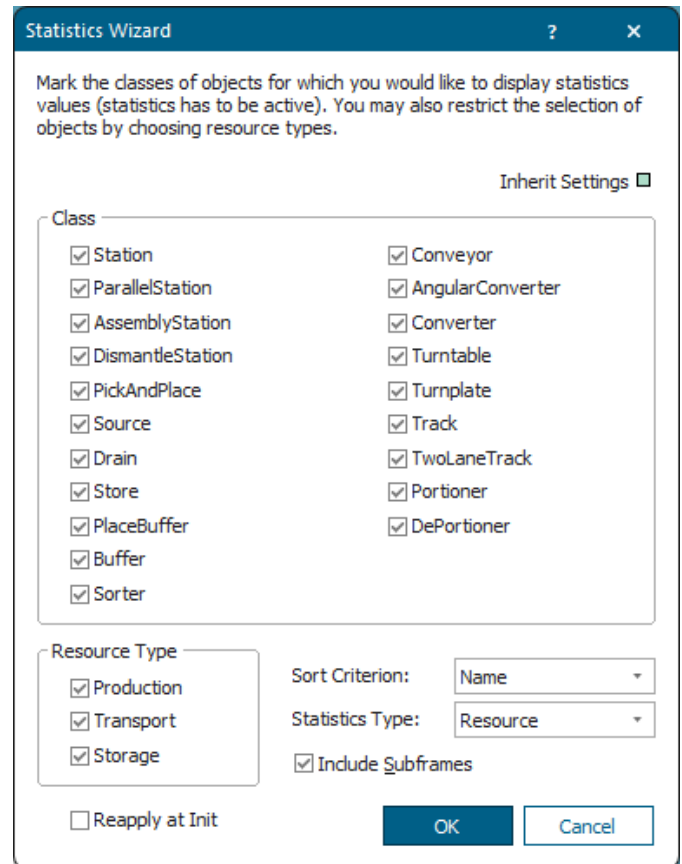
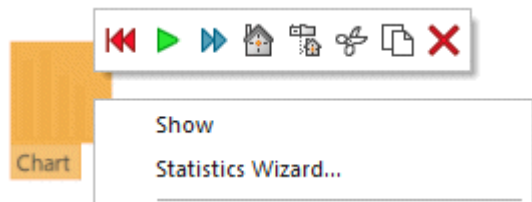


Description

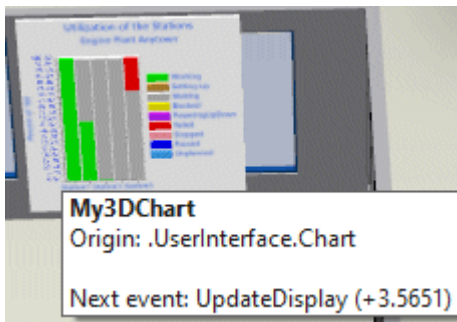
Define data to be plotted:

- By using a **Table** containing the data, such as simulation results.
- By defining **Input Channels** that record values of attributes of interest of the objects.

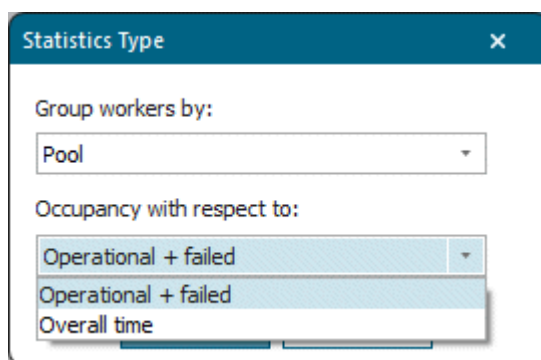
A *Chart*, which you insert into a *Frame*, provides the [Statistics Wizard](#). Select the command [Statistics Wizard](#) on the context menu of the *Frame* to open it.



To show a **tooltip** with information about the *Chart*, hover with the mouse over it.



To show statistics of the *Workers* managed by the *WorkerPool*, drag it over a *Chart* and drop it there. Select the settings that you need.

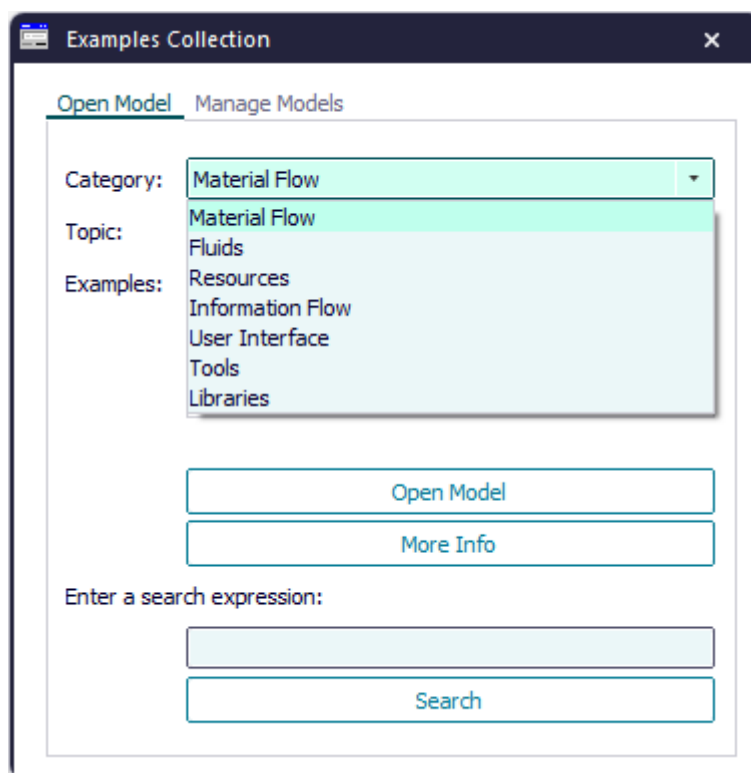


To change the length of the graphic and the anchor points of the *Chart*, click **Show Manipulators**  on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model

To add the object *Chart* to your simulation model, click  **Manage Class Library** > **Basic Objects** > **UserInterface** > **Chart** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page** > **Getting Started** > **Example Models** > **Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Show Statistics in a Chart](#)

[Statistics Wizard \[described\]](#)

[Drag-and-Drop in the Chart](#)

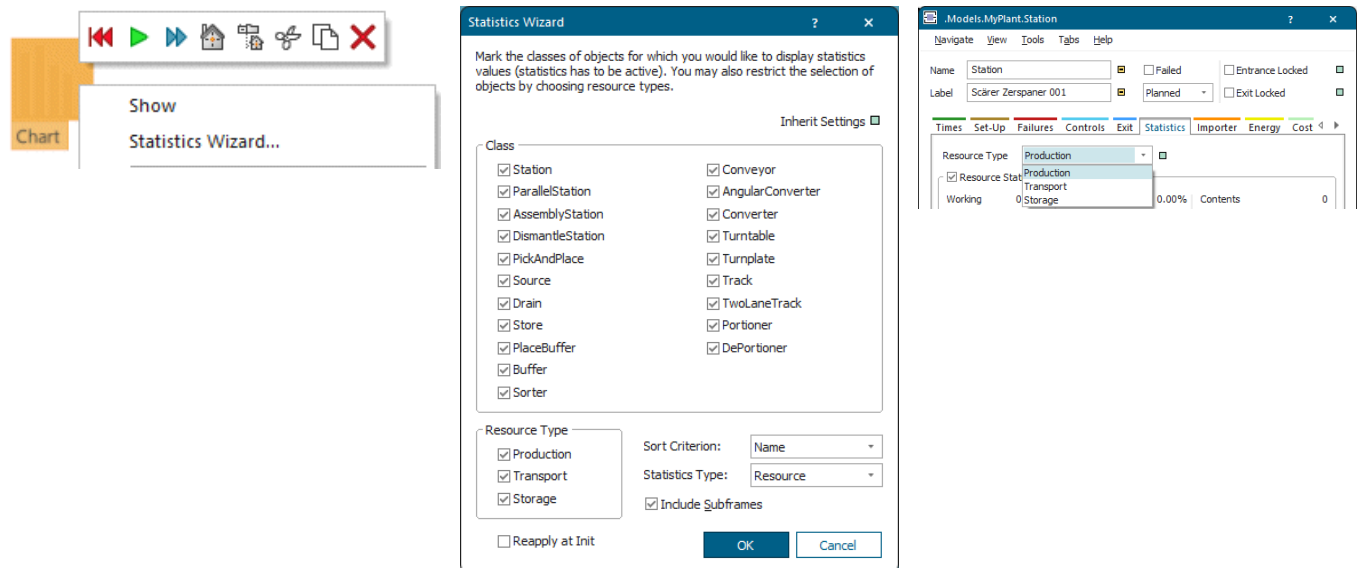
Statistics Wizard [described]

A *Chart*, which you insert into a *Frame*, shows the *Statistics Wizard*.

Remarks

Select the command **Statistics Wizard** on the context menu of the *Frame* to open it.

Select the class(es) of object(s) for which you would like to show statistics values in the *Chart*. The *Chart* only adds material flow objects to the display for which you selected the check box **Resource Statistics** on the **Tab Statistics**.



Note

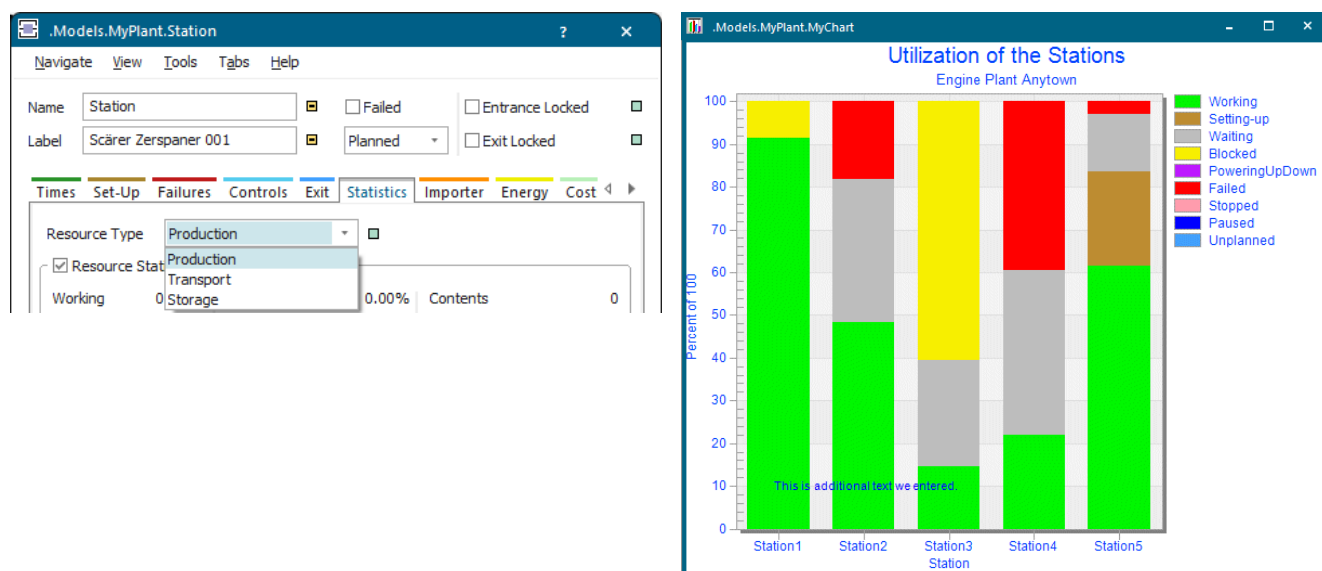
If you are working with a model that was created in a previous version of *Plant Simulation* that still uses the outdated object names of these versions, the dialog shows a **+ sign** in front of the new object names.

- To inherit the settings you select below, select **Inherit Settings** [Inherit settings](#) .
- Select the object classes in the group box **Class** for which you want to show statistics values in the *Chart*. The *Chart* only adds material flow objects to the display for which you selected the check box **Resource Statistics** on the **Tab Statistics**. You can select any number of **classes**.

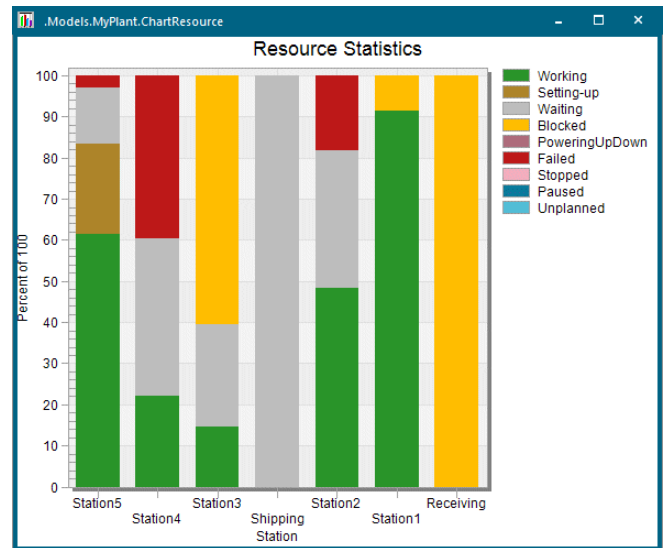
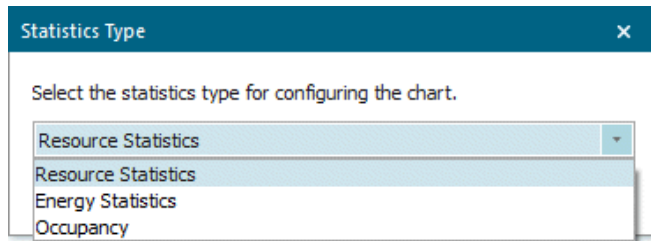
Select the type of resource(s) for which you would like to show statistical data of the selected object(s) in the *Chart* under **Resource Type** on the tab **Statistics** of an object collecting statistical data. With the check boxes in the *Statistics Wizard* you can restrict, which objects *Plant Simulation* shows.

- Production** is the default for the *Source*, *Drain*, *Station*, *ParallelStation*, *AssemblyStation*, and *DismantleStation*.

- **Transport** is the default for the *PlaceBuffer*, *Buffer*, *Sorter*, *Track*, *TwoLaneTrack*, *Conveyor*, *AngularConverter*, *Turntable*, *Transporter*, and *Container*.
- **Storage** is the default for the *Store*.
- Select the criteria according to which *Plant Simulation* sorts statistical data from the drop-down list **Sort Criterion**:
To sort the stations by their name in alphabetical order from A to Z, select **Sort Criterion > Name**.
To sort the stations in the order that meets your needs, drag the stations onto the *Chart* in the order it is to show them.
To change the order at a later point in time, edit the table of the Input channels by cutting the respective columns of the objects and pasting them in the desired order.
- Select the statistics type: **Resource Statistics**, **Energy Statistics**, or **Occupancy**.
- To also show statistical data of objects contained in *Frames* contained in the *Frame* in which the *Chart* is located, select **Include Subframes** ☒ *Include subframes*.
- To reapply the settings while the simulation model is initialized, select **Reapply at Init** ☐ *Reapply at Init*.
- Click **OK** to display the *Chart* window with the settings you selected.



- If you drag a material flow object onto a *Chart*, which you have not configured yet, you have to select the **statistics type**. By default, you can select **Resource** or **Occupancy**. If you activate **energy features**, you can also select **Energy**. The different statistics types provide different value sets. The meaning of the different states is described in the [Statistics Report](#) under [Resource Statistics of Stationary Resources](#) and [Energy Statistics — Energy Statistics](#).
- For the **Statistics type > Resource**, the *Chart* shows the **Name** of the object and the states it is in:
Working, **Setting-Up**, **Waiting**, **Blocked**, **Powering up/down**, **Failed**, **Stopped**, **Paused**, or **Unplanned**.

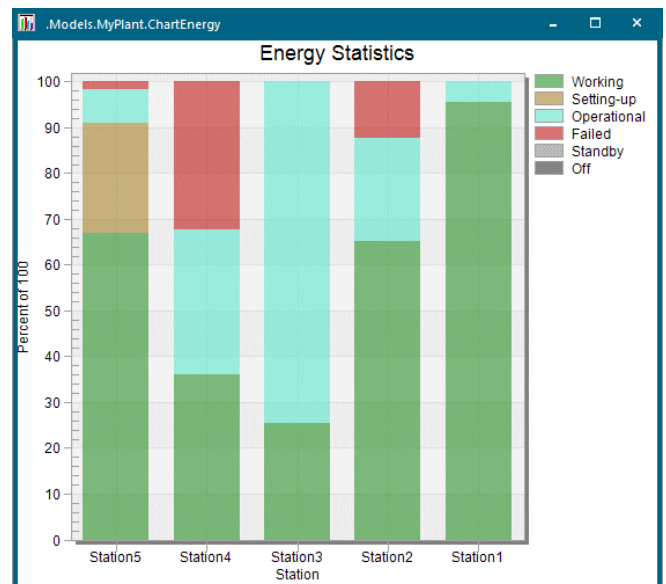
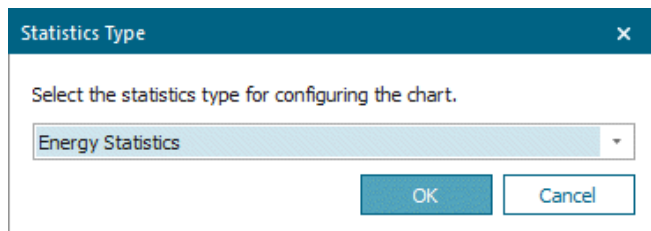


- For the **Statistics Type > Energy**, the *Chart* shows the energy states of the object:

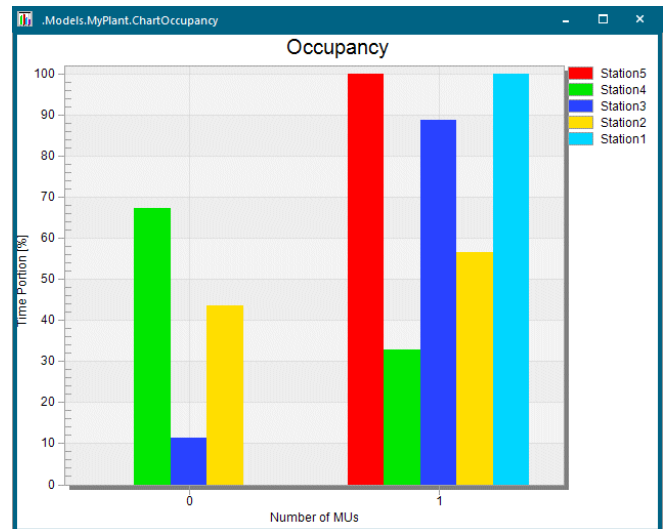
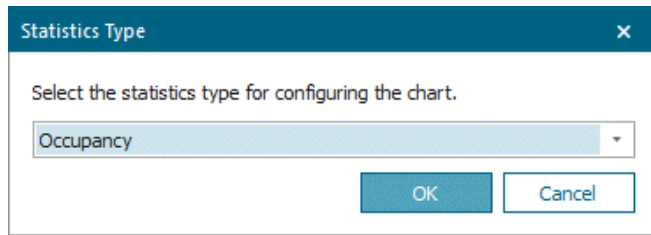
Working, Setting-up, Operational, Failed, Standby, and Off.

Note

The **energy states** of the material flow objects differ from the resource states with the same name. The values for the **resource states** refer to the **statistics collection period**, while the values for the **energy states** refer to the total **energy consumption**.



- For the **Statistics Type > Occupancy**, the *Chart* shows the **Name** of the object and the number of parts located on it. The colored bars serve for differentiating between the objects.



- You can also show the diagram picture of a *Chart* in an [HtmlReport](#), compare [Display a Chart](#).

Drag-and-Drop in the Chart

Drag-and-drop in the *Chart* performs a number of actions.

Remarks

Hold down the keys described below when dragging and dropping in the *Chart*.

To do this	Drag	To	Accelerator
Show the statistics of a single material flow object	material flow object	<i>Chart</i>	*
Show the statistics of multiple material flow objects	material flow objects	<i>Chart</i>	*
Show Driving statistics of the <i>Transporter</i>	<i>Transporter</i>	<i>Chart</i>	*
Show Services statistics of the <i>Worker/Exporter</i>	<i>Worker/Exporter</i>	<i>Chart</i>	*
Show Exporter statistics of the <i>Worker/Exporter</i>	<i>Worker/Exporter</i>	<i>Chart</i>	Shift
Show statistics of <i>Workers</i> controlled by a <i>WorkerPool</i>	<i>WorkerPool</i>	<i>Chart</i>	—
Show the frequency distribution of the number of MUs	<i>PlaceBuffer</i> , <i>Buffer</i> or <i>Store</i>	<i>Chart</i>	*
Show the frequency distribution for the individual part types of Supermarket	<i>Store</i> configured as Supermarket	<i>Chart</i>	—
Show statistics of all resources which are assigned to the <i>LockoutZone</i>	<i>LockoutZone</i>	<i>Chart</i>	*
Show the contents of a table	<i>DataTable</i> , <i>TimeSequence</i>	<i>Chart</i>	—
Show the value of the <i>Variable</i> you inserted into a <i>Frame</i>	<i>Variable</i>	<i>Chart</i>	*
Show statistics of a <i>Frame</i> , which has the user-defined attribute <i>StatisticsObject</i>	User-defined object (<i>Frame</i>)	<i>Chart</i>	*
Show statistics of the object <i>Workplace</i> within a <i>Frame</i> , which has the user-defined attribute <i>AOLType</i>	User-defined object (<i>Frame</i>)	<i>Chart</i>	*

* If you hold down **Shift** while dropping the object onto the *Chart*, *Plant Simulation* does not add the dropped object to the **Data Source > Input Channels**, but deletes the existing entries from the input channels.

The method **addObject** accomplishes the same as using drag-and-drop actions described above.

Note

To delete one of the displayed objects, click **Data Table** next to **Data Source** and delete the object from the data table.

To delete the entire contents of the input channels table, hold down **Shift** while you drag-and-drop a new object onto the *Chart*.

If you want your own object, which you modeled in a *Frame*, to behave like one of the built-in *Plant Simulation* objects when you drag-and-drop it, create a *user-defined attribute* of data type *object* for this *Frame* and name it `StatisticsObject`. This attribute may designate any of the material flow objects. The *Chart* then acts as if the object, to which the *user-defined attribute* refers, was dropped onto the *Chart*. This way you can show the statistics of a *Frame* according to your needs.

If your *Frame* contains a number of objects whose statistics you would like to show, then assign the data type *table* to the *user-defined attribute* named `StatisticsObject`. The first column of the table has to be of data type *string*. The *Chart* now acts if all objects contained in the first column of the table were dropped onto the *Chart*. You can also use a user-defined row index and type in the labels of the input channels.

Dialog Box of the Chart

Dialog Box of the Chart

Double-click the icon of the *Chart* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

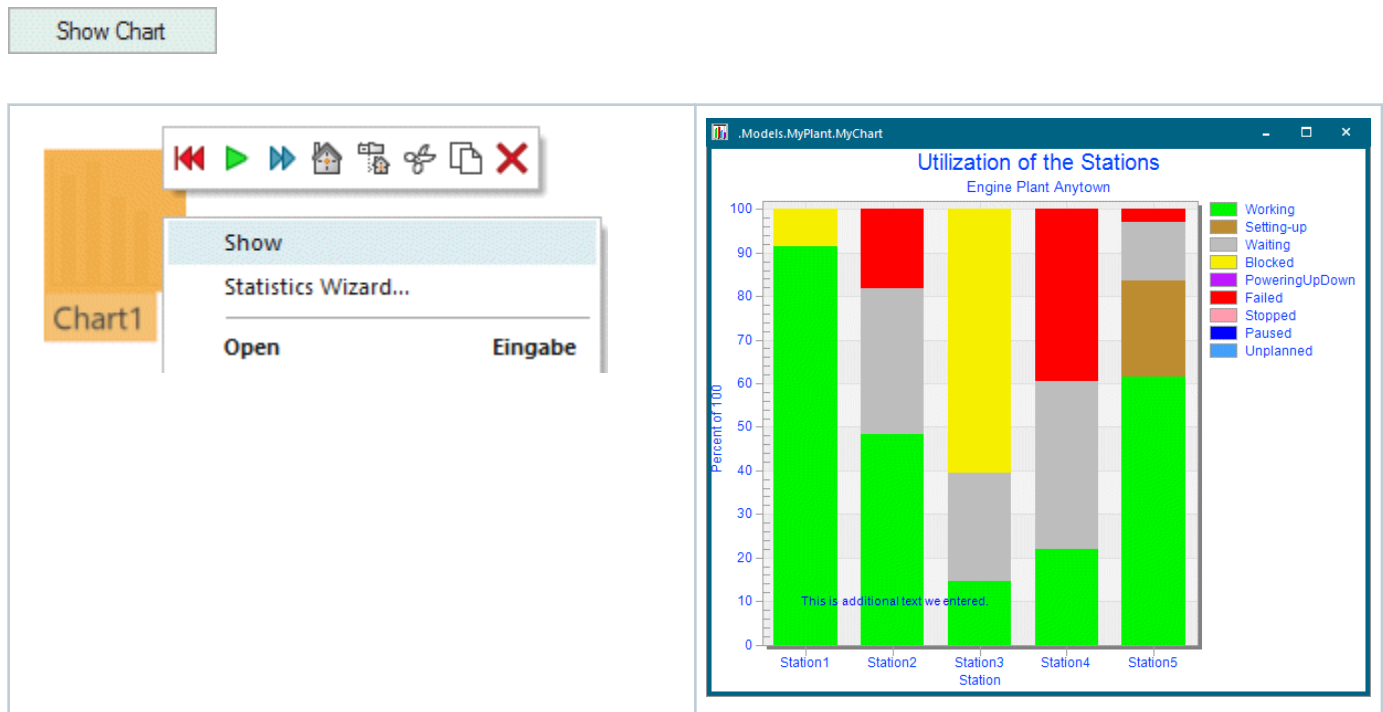
See also

[Show Chart \[button\] - Chart](#)

[Collect Data \[check box\] - Chart](#)

Show Chart [button] - Chart

To show the data, which the *Chart* collected in the display window, click this button.



Remarks

To open the dialog of the *Chart*, double-click anywhere in the display window.

To delete one of the displayed objects, click **DataTable** next to **Data Source** and delete the object from the input channels table.

To delete the entire contents of the input channels table, hold down **Shift** while you drag-and-drop a new object onto the *Chart*.

To show the data, you can also right-click the *Chart* object in the *Frame* and select **Show** on the context menu.

If your *Chart* contains a large number of values, you can zoom an area of interest by dragging a selection rectangle over the area.

Accelerators for the Chart

You can press the following keys on the keyboard in the display window of the *Chart* to execute these actions:

To do this	Press this Keyboard Key
Switch between monochrome and color viewing styles	S
Show the Export dialog	X
Maximize the window to full screen	M
Zoom all, i.e., to show all data without a scrollbar.	Z
Show the Text/Data export dialog.	D
Show the Print dialog	P
Scroll both vertical and horizontal scrollbars by one line.	Arrow keys
Page vertical scrollbar.	PgUp/PgDown
Page horizontal scrollbar to the right.	Shift+PgUp
Page horizontal scrollbar to the left.	Shift+PgDown
Move vertical scrollbar to its first position.	Home
Move vertical scrollbar to its last position.	End
Move horizontal scrollbar to its left-most position.	Shift+Home
Move horizontal scrollbar to its right-most position.	Shift+End

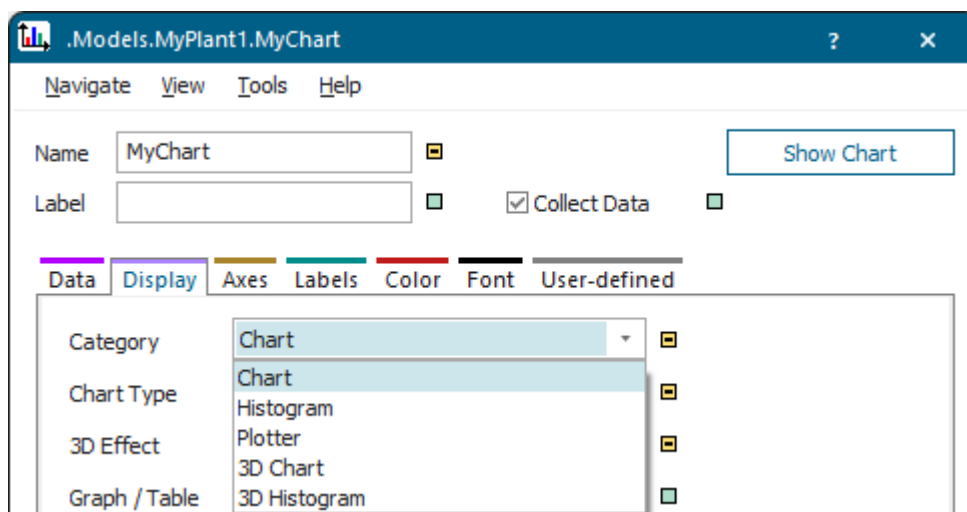
SimTalk

IsShown [SimTalk] - Chart

Collect Data [check box] - Chart

To make a *Chart* of the **Category > Histogram** or **Plotter** collect data during the simulation run, select this check box. To deactivate data collection, clear the check box.

☒ Collect Data ☐



SimTalk

CollectData [SimTalk] - Chart

See also

Category [drop-down list]

Tab Data

Tab Data

On the tab **Data** you can select the **Data Source**, i.e., which data the *Chart* displays, from the drop-down list.

Data	Display	Axes	Labels	Color	Font	User-defined
Data Source	Data Table					☐
Table	MyDataTable					☐
Range	{1,1}..{*,*}					☐
Data	in Row					☐
Mode	Sample					☐
Interval	1:00					☐

Remarks

You can select one of these settings:

- Data from a **Data Table**.
- Data from **Input Channels**.

You can also select the **Mode**:

- **Sample** mode shows sampled data after an interval you enter.
- **Watch** mode updates the data each and every time one of the displayed values changes.
- **Plot** mode updates the display whenever a simulation event takes place.

In addition, you can select different views on the **Data** to be displayed.

See also

[Data Source \[Chart\]](#)

Data Source [Chart]

Select from where the *Chart* gets the data you want to display.

Data Table

For this setting the *Chart* shows the contents of a *DataTable* as a diagram.

The screenshot shows the 'Data Source [Chart]' dialog box with the 'Data Table' tab selected. The dialog has seven tabs: Data, Display, Axes, Labels, Color, Font, and User-defined. The 'Data' tab is active, showing the following settings:

- Data Source:** Data Table (dropdown menu)
- Table:** MyDataTable (text field with a selection button '...')
- Range:** {1,1}..{*,*} (text field)
- Data:** in Row (dropdown menu)
- Mode:** Sample (dropdown menu)
- Interval:** 1:00 (text field)

Each setting has a small icon to its right: a square with a minus sign for dropdowns and a square with a plus sign for text fields.

Type in the name of and path to the data table containing the data, or click the button and select the table in the dialog [Select Object](#).

Type in the **Range** of the table cells you want to show.

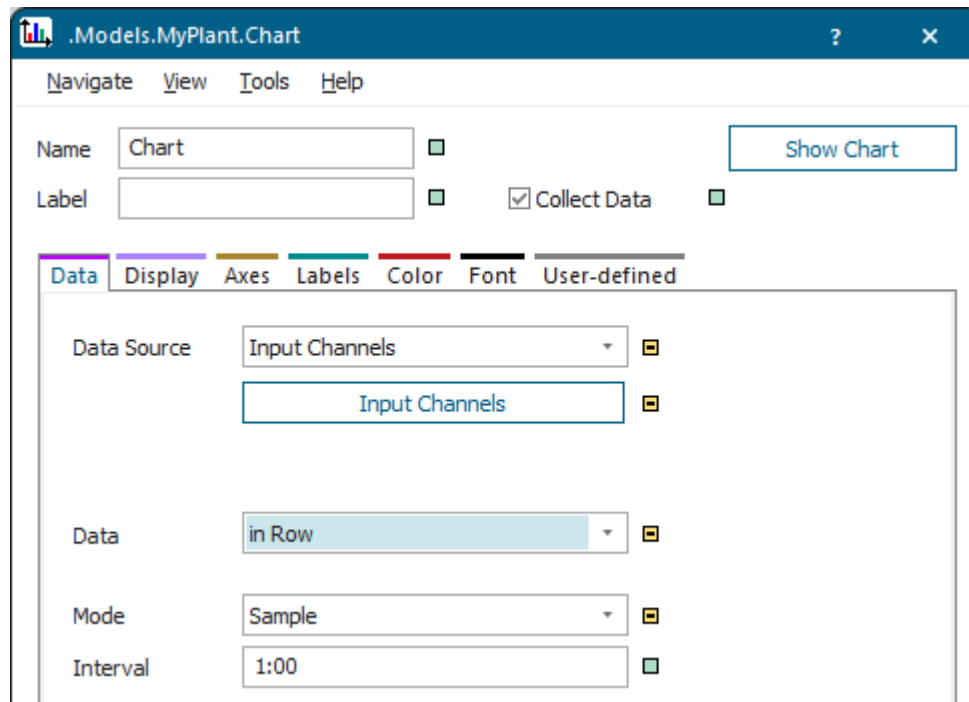
- To show all cells in all columns and rows, type in $\{*, *\}.. \{*, *\}$.
- To show the first four cells of column one, type in $\{1, 1\}.. \{1, 4\}$.
- To show all cells in column one, type in $\{1, 1\}.. \{1, *\}$.

You can also type in negative numbers:

- To show the one but last column, type in $\{-2, *\}.. \{-2, *\}$.
- To show the one but last and the last column of the table, type in $\{-2, *\}.. \{*, *\}$.

Input Channels

For this setting the *Chart* shows the channels contained in the **Input Channels** table.



To open a table into which you type in the input channels, click **Input Channels**.

These channels are, as a rule, the paths to attributes or methods of objects, whose values you want to display. You can also type in complex expressions, i.e., method calls, formulas, or the sum of two attributes. Also type in row and column headers as labels of the *Chart*.

Station1.StatWorkingPortion				
	string 0	string 1	string 2	string 3
string		Station1	Station2	Station3
1	Working	Station1.StatWorkingPortion	Station2.StatWorkingPortion	Station3.StatWorkingPortion
2	Setting-up	Station1.StatSetupPortion	Station2.StatSetupPortion	Station3.StatSetupPortion
3	Waiting	Station1.StatWaitingPortion	Station2.StatWaitingPortion	Station3.StatWaitingPortion
4	Blocked	Station1.StatBlockingPortion	Station2.StatBlockingPortion	Station3.StatBlockingPortion
5	PoweringUpDown	Station1.StatPoweringUpDownPortion	Station2.StatPoweringUpDownPortion	Station3.StatPoweringUpDownPortion
6	Failed	Station1.StatFailPortion	Station2.StatFailPortion	Station3.StatFailPortion
7	Stopped	Station1.StatStoppedPortion	Station2.StatStoppedPortion	Station3.StatStoppedPortion
8	Paused	Station1.StatPausingPortion	Station2.StatPausingPortion	Station3.StatPausingPortion
9	Unplanned	Station1.StatUnplannedPortion	Station2.StatUnplannedPortion	Station3.StatUnplannedPortion

To delete the entire contents of the input channels table, hold down **Shift** while you drag-and-drop a new object onto the *Chart*.

Note

To specify a formula, type it directly into the respective cell of the *DataTable*.

Do not click the formula button  on the **List** ribbon tab of the *DataTable*!

Note

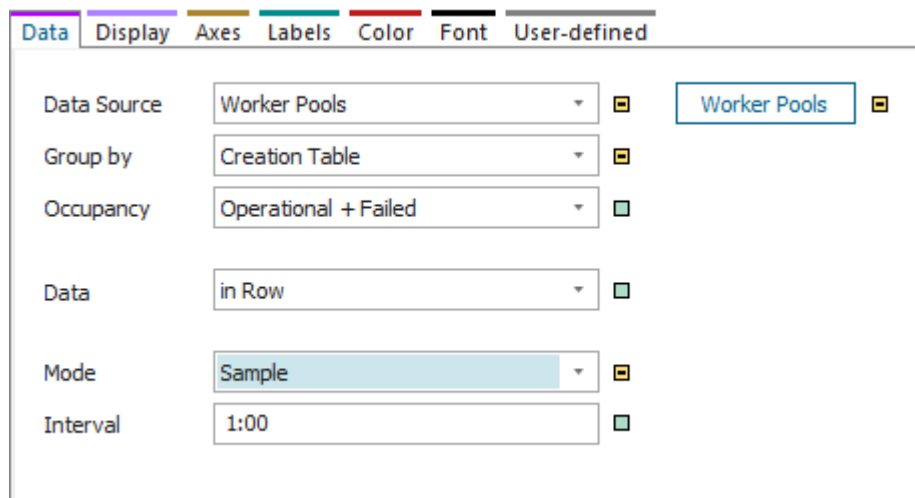
To change the order in which the *Chart* shows the stations, edit the **Table** of the **Input Channels** by cutting the respective columns and pasting them in the desired order.

Note

If the amount of data to be displayed is extremely large, it can take some time to show the *Chart*. This applies, for example, to a table containing several tens of thousands of cells. Some chart types, for example the 3D types, also take a comparatively large amount of time to display.

Worker Pools

For this setting the *Chart* shows the *Workers* managed by *WorkerPools* as a diagram.



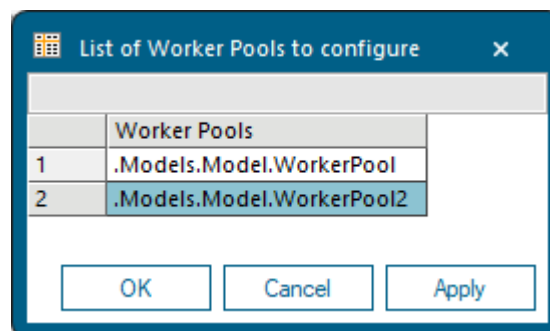
The dialog box has tabs: Data, Display, Axes, Labels, Color, Font, and User-defined. The 'Data' tab is active. It contains the following settings:

- Data Source: Worker Pools
- Group by: Creation Table
- Occupancy: Operational + Failed
- Data: in Row
- Mode: Sample
- Interval: 1:00

There are checkboxes to the right of each setting, and a 'Worker Pools' button with a dropdown arrow.

- To open a table into which you type in the *WorkerPools* to be used, click **Worker Pools**.

Drag the *WorkerPools* whose *Workers* you want to show to the cells of the dialog **List of Worker Pools to Configure** and click OK.



The dialog box has a title bar 'List of Worker Pools to configure' and a close button. It contains a table with the following data:

	Worker Pools
1	.Models.Model.WorkerPool
2	.Models.Model.WorkerPool2

At the bottom, there are buttons for OK, Cancel, and Apply.

- Select how you want to **group** the *Workers*. According to the:
 - **Pool**, which uses the entire *WorkerPool*.
 - **Creation Table**, which only uses the *Workers* contained in the **Workers to Create** table of the *WorkerPool*.

The screenshot shows the 'Data' tab of a configuration window. The 'Data Source' is set to 'Worker Pools'. The 'Group by' dropdown menu is open, with 'Creation Table' selected. The 'Occupancy' dropdown menu is also open, showing 'Pool' and 'Creation Table' as options. Other settings include 'Data' set to 'in Row', 'Mode' set to 'Sample', and 'Interval' set to '1:00'.

- Select which statistics values for the **Occupancy** you want to show:
 - **Operational + failed** shows the time during with the *Worker* was **operational** plus the time during which he was **failed**.
 - **Overall Time**

The screenshot shows the 'Data' tab of a configuration window. The 'Data Source' is set to 'Worker Pools'. The 'Group by' is set to 'Pool'. The 'Occupancy' dropdown menu is open, with 'Operational + failed' selected. The 'Data' dropdown menu is also open, showing 'Operational + failed' and 'Overall Time' as options. The 'Mode' is set to 'Watch'.

Note

To initially add the *WorkerPools* to the *Chart*, you can select them, drag them over the icon of the *Chart* and drop them there.

The screenshot shows the 'Statistics Type' dialog box. The 'Group Workers By' dropdown is set to 'Pool'. The 'Occupancy With Respect To' dropdown is set to 'Operational + Failed'. There are 'OK' and 'Cancel' buttons at the bottom.

Then continue configuring the *Chart* on the tab **Data**.

SimTalk

[DataRange \[SimTalk\]](#)

[DataTable \[SimTalk\] - Chart](#)

[InputChannels \[SimTalk\]](#)

[UseInputChannels \[SimTalk\]](#)

See also

[Show Worker Statistics in a Chart](#)

Data [Chart]

Select how the *Chart* shows the data to be visualized.

Data

in Row

in Row

in Column

in Column

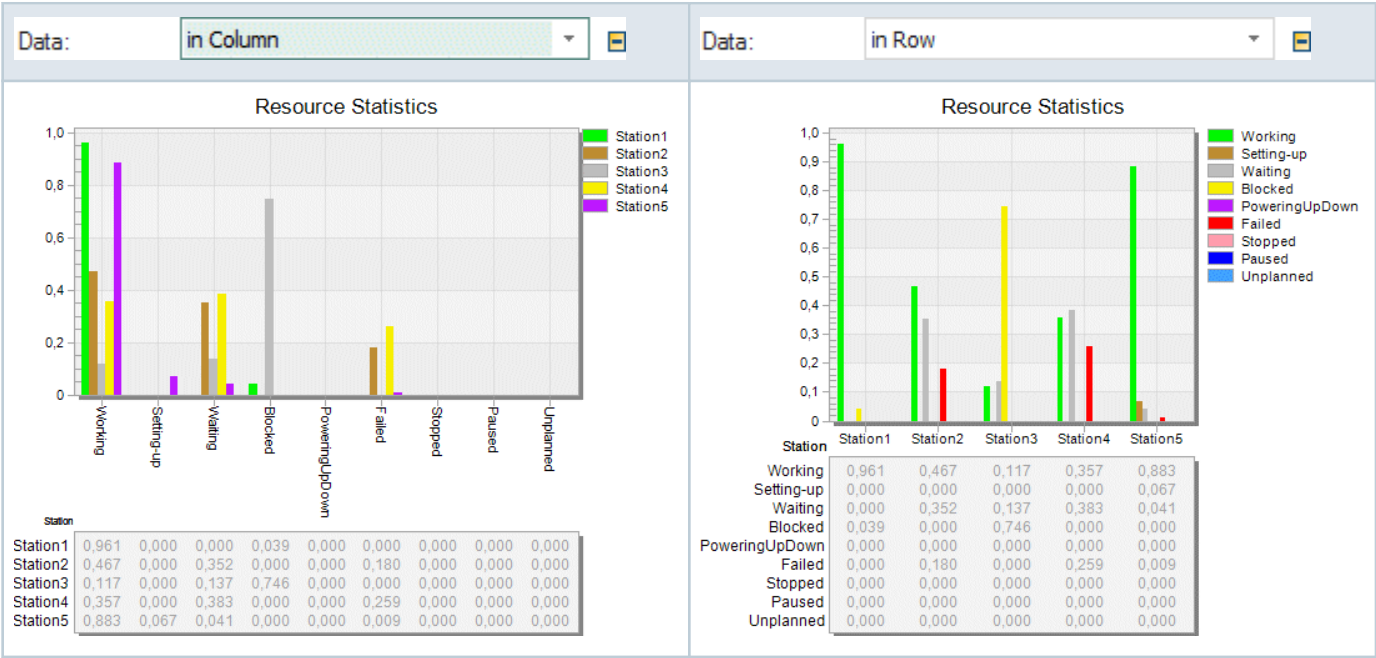
Mode

Remarks

You can select one of these settings:

- In Rows
- In Columns

This way you can present different views of the same record.



SimTalk

DataInColumn [SimTalk]

Mode [drop-down list] - Chart

Select the **Mode** with which the *Chart* updates the displayed data.

Mode	Watch Watch Sample Plot	Mode	Sample
		Interval	1:00

Remarks

You can select one of these settings:

- Sample Mode**

Updates the *Chart* periodically after the **interval** you enter.

Type in the time span (**Interval**) after which the *Chart* takes the next sample.

- Watch Mode**

Updates the *Chart* whenever the **watchable** input data changes.

The column **Watchable** in the window **Show Attributes and Methods**, shows the *methods* and *attributes* which *Plant Simulation* can watch.

Name	Value	Inherited	Watchable	Signature
A2Ctrl		-		method
addObserver				(AttributeName:string, Method:method)
AssignedLockoutZones	[]			-> array
assignedWorkplaces	[]			([Workplaces:table]) -> any
attributeWatchable				(AttributeName:string) -> boolean
AutomaticProcessing	true	✓	*	boolean
AutomaticSetup	true	✓	*	boolean
Availability	100.0			real
Capacity	1	✓	*	integer
ChangePathCtrl	VOID	✓		method
childNo				(No:integer) -> object

- Plot Mode**

Updates the *Chart* whenever a simulation event takes place.

SimTalk

update [SimTalk] - Chart]

[NumIntervals \[SimTalk\]](#)

[Mode \[SimTalk\] - Chart](#)

[SampleInterval \[SimTalk\]](#)

Tab Display

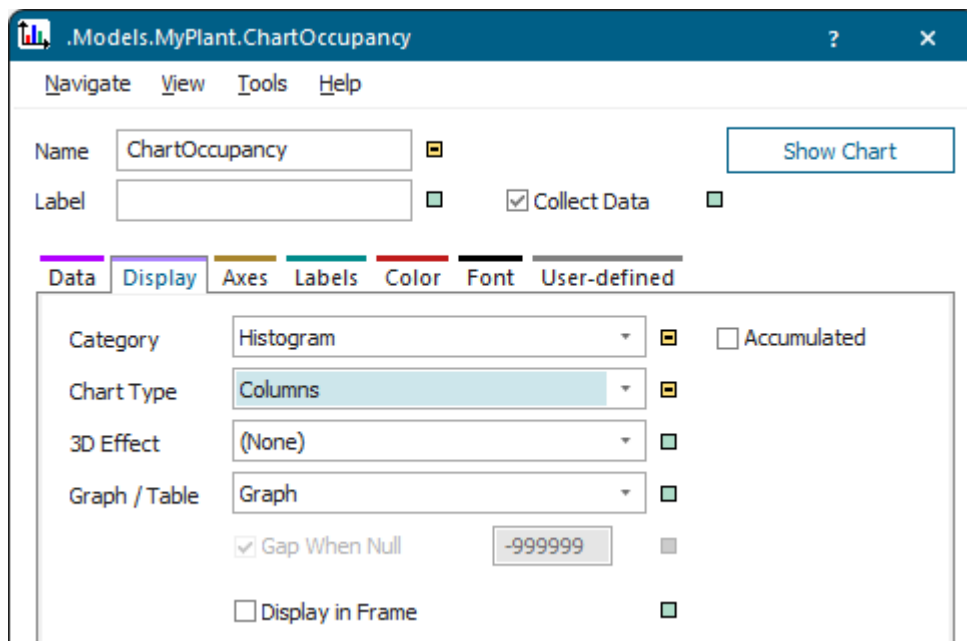
Tab Display

Select how the *Chart* shows the data on the tab **Display**.

Remarks

The tab **Display** provides the settings listed in the table of contents to the left.

- For the **Category** > **Histogram**, you can show the data [Accumulated](#).



- For the **Category** > **Plotter**, you can show the data as a [Step Curve](#).

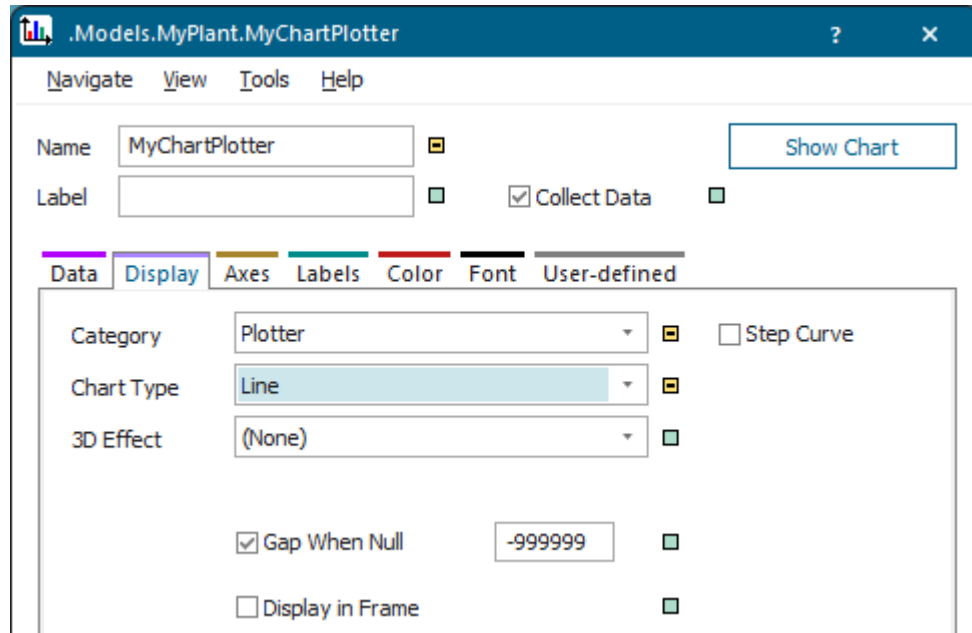


Chart Category

Category [drop-down list]

Select the **Category** for displaying the data.

Category:

Remarks

You can select one of these settings:

- **Chart**
Shows a number of values or several sets of a number of values.
- **Histogram**
Shows the frequency distribution for one or several input channels. For this setting you can select the check box **Accumulated**.

Note

The input channels for **Histograms** should to be **watchable** to show exact values.

- **Plotter**
Shows the progression of one or more values over time. For this setting you can select the check box **Step Curve**.
- **3D Chart**

Shows a **3D Chart** embedded in the 3D display window with the settings you select. You cannot open the **3D Chart** in a window of its own by clicking **Show Chart**.

- **3D Histogram**

Shows a **3D Histogram** embedded in the 3D display window with the settings you select. You cannot open the **3D Histogram** in a window of its own by clicking **Show Chart**.

- **XY Graph**

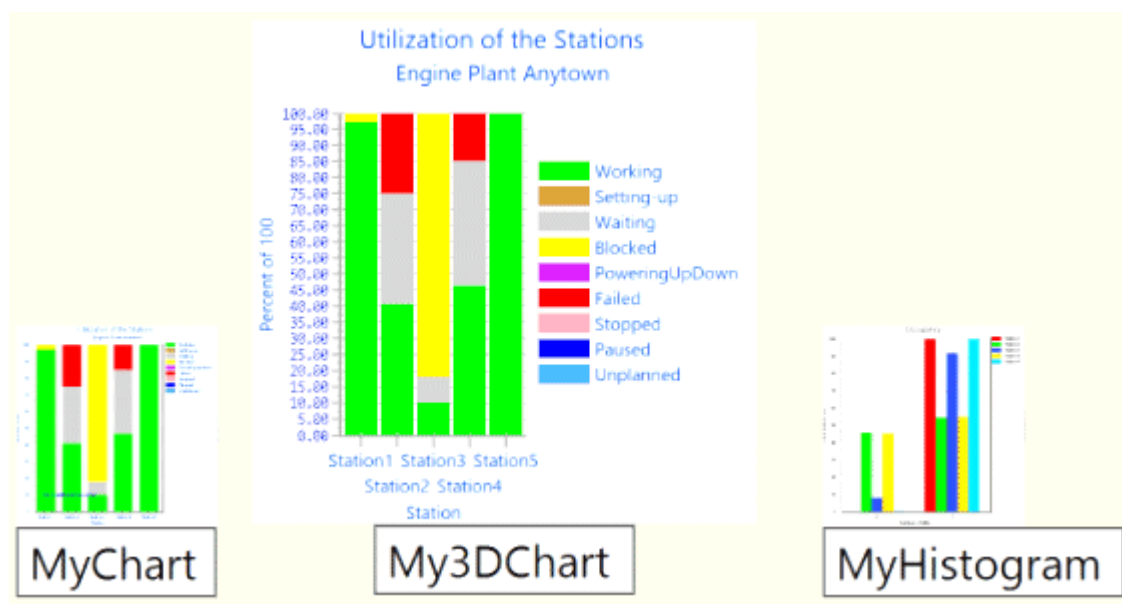
Shows a number of x-y pairs of values or several sets of x-y pairs of values. This setting is only active if you select **Table File** as the **Data Source**.

The **XY Graph** shows the range like this:

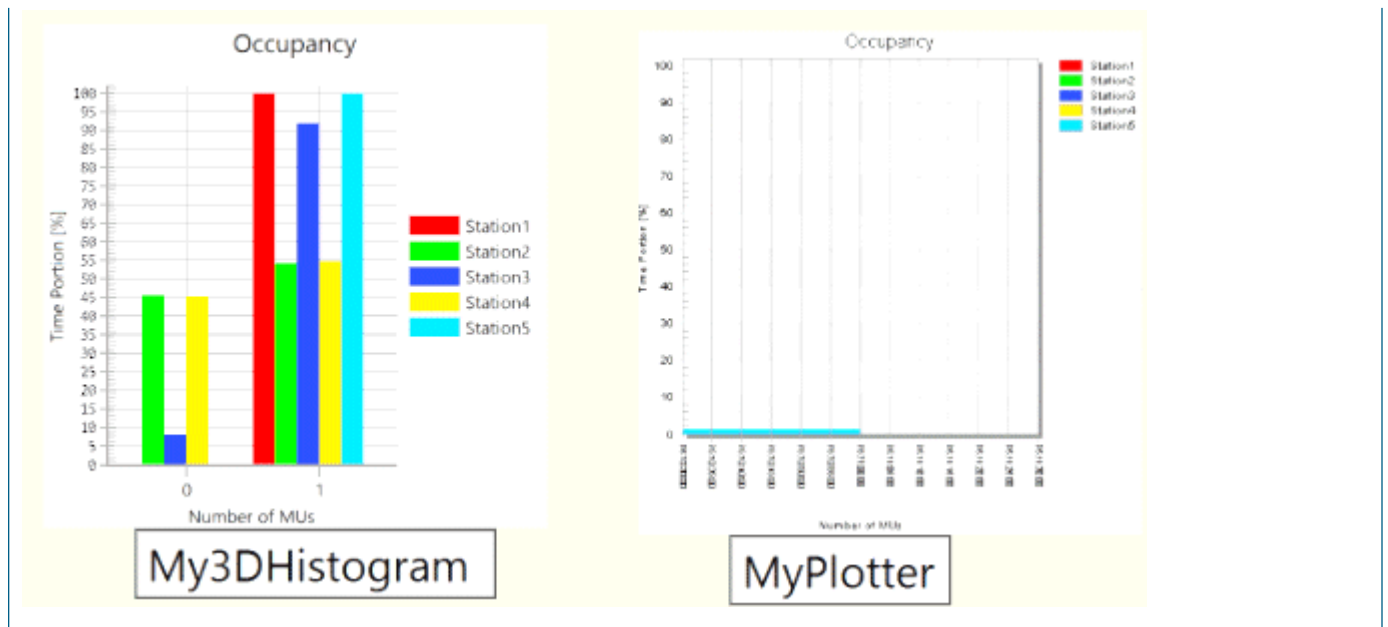
- The x values are contained in the first row or column.
- The range determines the y values. The first row or column, which contains the x values, will automatically be excluded.

Note

If you select another **Category**, the *Chart* in 3D automatically shows the graphic of this category in the *Frame*.



Category [drop-down list]



SimTalk

Category [SimTalk]

putValuesIntoTable [SimTalk] - Chart

resetValues [SimTalk]

Category - 3D Chart

The **Category 3D Chart** shows a **3D Chart** embedded in the 3D display window with the settings you select.

Category	3D Chart		Category	3D Chart	
Chart Type	100% Stacked Columns		Chart Type	100% Stacked Columns	
3D Effect	Columns		3D Effect	3D	
Graph / Table	100% Stacked Columns		Graph / Table	Graph	
	Area		☒ Gap When Null	-999999	
	Stacked Area		☒ Display in Frame		
	100% Stacked Area		Width	5	Height
	Line				5 m

Remarks

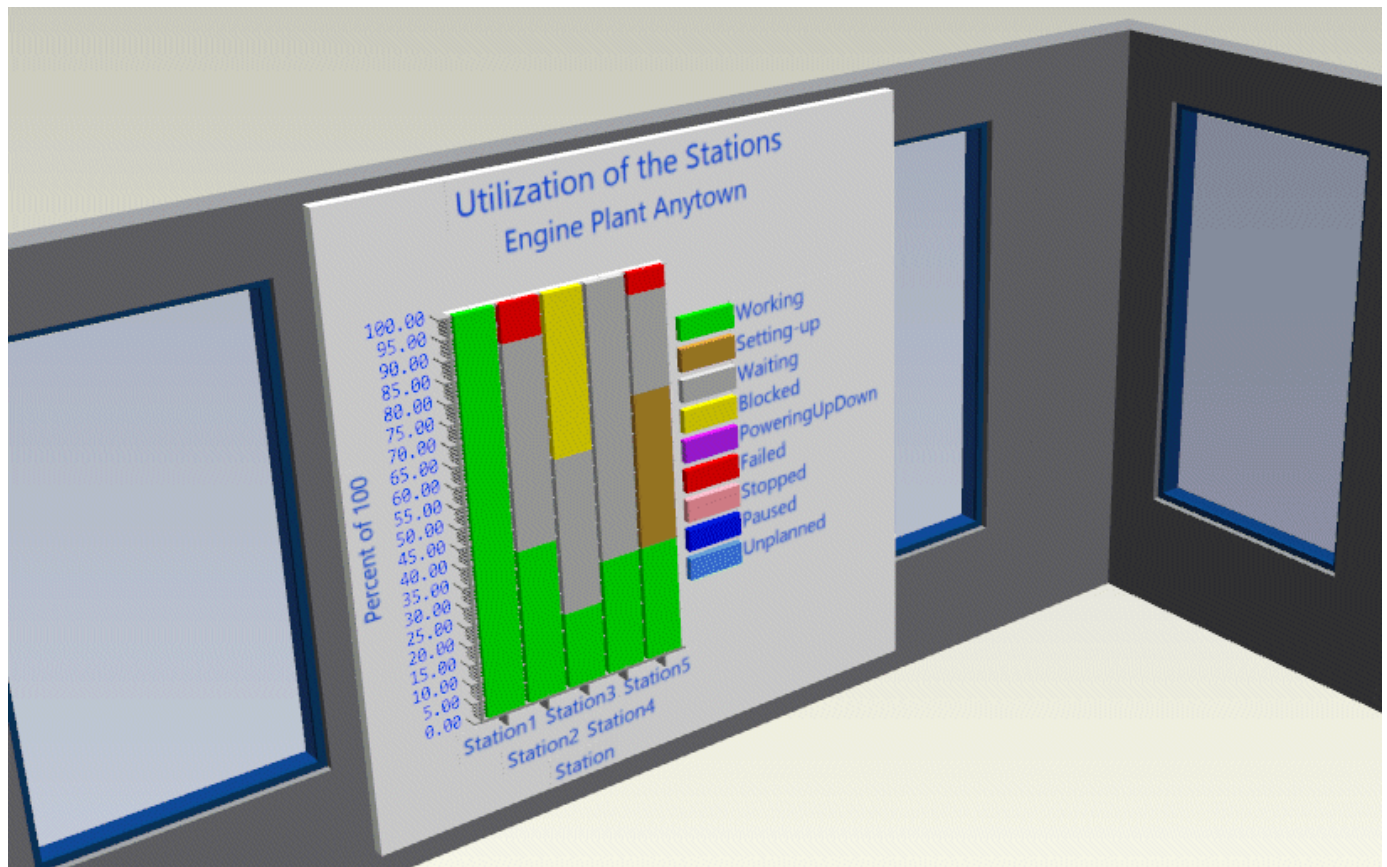
Note

You cannot open the **3D Chart** in a window of its own by clicking **Show Chart**.

The category **3D Chart** has these properties:

- It has a fixed **Font Size**. Only the setting **Bold** affects it. For a **Width** and/or **Height** greater than 5 meters, the font size is absolutely fixed. For a size smaller than 5 meters the font size scales down with all other display elements.
- It uses the settings for the **Color** and the **Line Weight**, which you select on the **Tab Color**, for the setting **Chart Type > Line**. It does neither use a **Marker Type** nor a **Line Style**.
- It enables you to hide the **Base plate** and the **background of the displayed data** by setting the **Opacity** of the **Background** and the **Desk** to 0 on the **Tab Color**. This way you can create a transparent chart which you can lay on top of other graphics or objects.
- Its **Base plate** has a thickness of 20 cm by default. If you set the **Opacity** of the **Background** to 0, it has a thickness of 10 cm. For a **Width** and/or **Height** smaller than 5 meters, the thickness scales down with all other display elements. If the minimum size is 2.5 m for example, the chart with a base plate has a thickness of 10 cm and a thickness of 5 cm without a base plate.

A **3D Chart** attached to the wall of your factory might look like this.



See also

Category [drop-down list]

Category - 3D Histogram

The **Category 3D Histogram** shows a **3D Histogram** embedded in the 3D display window with the settings you select.

Category: 3D Histogram

Chart Type: Chart, Histogram, Plotter, 3D Chart, 3D Histogram

3D Effect: [checkbox]

Graph / Table: [checkbox]

☒ Gap When Null -999999

☒ Display in Frame

Width: 2 Height: 2 m

Remarks

Note

You cannot open the **3D Histogram** in a window of its own by clicking **Show Histogram**.

The category **3D Histogram** has these properties:

- It has a fixed **Font Size**. Only for the setting **bold** affects it. For a **Width** and/or **Height** greater than 5 meters, the font size is absolutely fixed. For a size smaller than 5 meters the font size scales down with all other display elements.
- It uses the settings for the **Color** and the **Line Weight**, which you select on the **Tab Color**, for the setting **Chart Type > Line**. It does neither use a **Marker Type** nor a **Line Style**.
- It enables you to hide the **Base plate** and the **background of the displayed data** by setting the **Opacity** of the **Background** and the **Desk** to 0 on the **Tab Color**. This way you can create a transparent histogram which you can lay on top of other graphics or objects.
- Its **Base plate** has a thickness of 20 cm by default. If you set the **Opacity** of the **Background** to 0, it has a thickness of 10 cm. For a **Width** and/or **Height** smaller than 5 meters, the thickness scales down with all other display elements. If the minimum size is 2.5 m for example, the chart with a base plate has a thickness of 10 cm and a thickness of 5 cm without a base plate.

See also

[Category \[drop-down list\]](#)

Chart Type [drop-down list]

Select the **Type** of *Chart* which displays the values of the simulation run. A record consists of the entries of a row or a column, depending on the setting you selected under **Display > Data**.

Chart Type 100% Stacked Columns 

Remarks

Depending on the **Category** you selected, the *Chart* provides some or all of these chart types:

Category	Description
Columns	Shows each record as a column. When displaying several records, the <i>Chart</i> places the columns next to each other. Use this to compare values across categories.
Stacked Columns	Shows each record as a column. When displaying several records, the <i>Chart</i> places the columns of the individual records on top of each other. It stacks positive values upward, negative values downward.
100% Stacked Columns	Shows each record as a column, stacked one on top of the other, showing the respective percentage of the entire set of 100 percent.
Bars	Shows each record as bars positioned horizontally on top of each other. When displaying several records, the <i>Chart</i> places the bars of the individual records next to each other. Use this to compare values across categories.
Stacked Bars	Shows each record as a set of bars stacked horizontally next to each other. Positive values are stacked to the right, negative values are stacked to the left.
100% Stacked Bars	Shows each record as a set of bars, one stacked horizontally next to each other, showing its respective percentage of the entire set of 100 percent.
Area	Shows each record as areas. When displaying several records, the <i>Chart</i> stacks the areas from the individual records one behind the other. Use this to show the trend of the contribution of each value over time or categories.
Stacked Area	Shows each record as areas. When displaying several records, the <i>Chart</i> stacks the areas from the individual records one on top of the other.
100% Stacked Area	Shows each record as a set of areas, one stacked on top of the other, showing the respective percentage of the entire set of 100 percent. Displays the trend of the percentage each value contributes over time or categories.
Line	Shows each record as a line.
Line with Markers	Shows each record as a line with markers.
Spline	Shows each record as a curve.
Spline with Markers	Shows each record as a curve with markers.

Category	Description
Markers	Shows each record as a set of different markers.
Points and Best Fit Conveyor	Shows each record as a set of data points, and fits the best straight line to them. The line is calculated from least-squares-approximations of the data set.
Points and Best Fit Curve	Shows each record as a set of data points, and fits the best curved line to them. The curve is calculated from least-squares-approximations of the data set.
Pie	Shows each record as a slice of a pie, depicting its percentage of the whole. Enter a negative prefix, so that the <i>Chart</i> offsets this slice from the rest of the pie. Use the scrollbar to scroll from record to record. Use this to show the contribution of each value to a total.
XY Points	Shows x-y pairs as data points. The first record contains the x-coordinates. The <i>Chart</i> interprets all additional records as the associated y-coordinates. Use this to compare pairs of values.
Line	Shows x-y pairs as data points and connects them with a line. The first record contains the x-coordinates. The <i>Chart</i> interprets all additional records as the associated y-coordinates.
Sticks	Shows a vertical bar for each x-y pair, starting from the zero line. The first record contains the x-coordinates. The <i>Chart</i> interprets all additional records as the associated y-coordinates.
Area	Shows the connected points belonging to the x-y pairs and fills the areas thus created. The first record contains the x-coordinates. The <i>Chart</i> interprets all additional records as the associated y-coordinates.
3D Columns	Shows each record as columns stacked in three-dimensional space. To turn the chart left or right you can enter an angle of Rotation or you can also drag the horizontal slider in the display window. To turn the chart up or down, you can enter the Height or you can drag the vertical slider.
3D Wire Frame	Shows each record as wire-frames stacked in three-dimensional space. To turn the chart left or right you can enter an angle of Rotation or you can also drag the horizontal slider in the display window. To turn the chart up or down, you can enter and the Height or you can drag the vertical slider.
3D Surface	Shows each record as solid areas stacked in three-dimensional space. To turn the chart left or right you can enter an angle of Rotation or you can also drag the horizontal slider in the display window. To turn the chart up or down, you can enter the Height or you can drag the vertical slider.

SimTalk

[ChartType \[SimTalk\]](#)

[Rotation \[SimTalk\] - Chart](#)

Chart Type [drop-down list]

Height [SimTalk] - 3D charts

See also

Data [Chart]

3D Effect

Select one of the **3D effects**.

3D Effect (None) ☐

Graph / Table (None) ☐

Shadow ☐

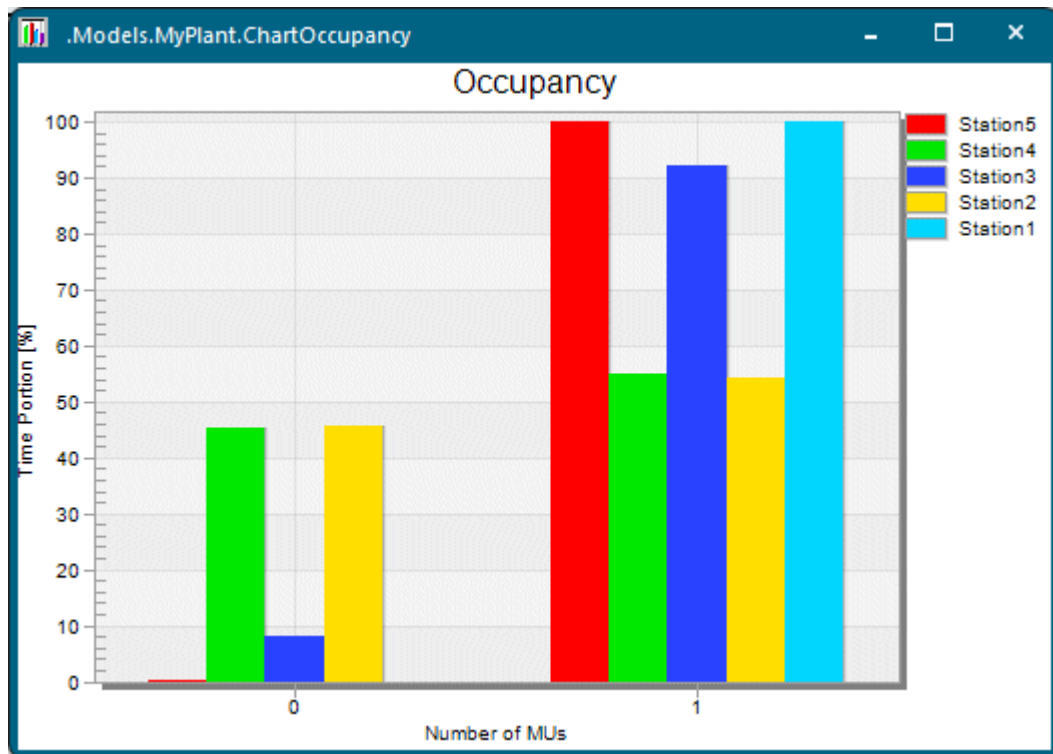
3D ☐

Gradient bars ☐

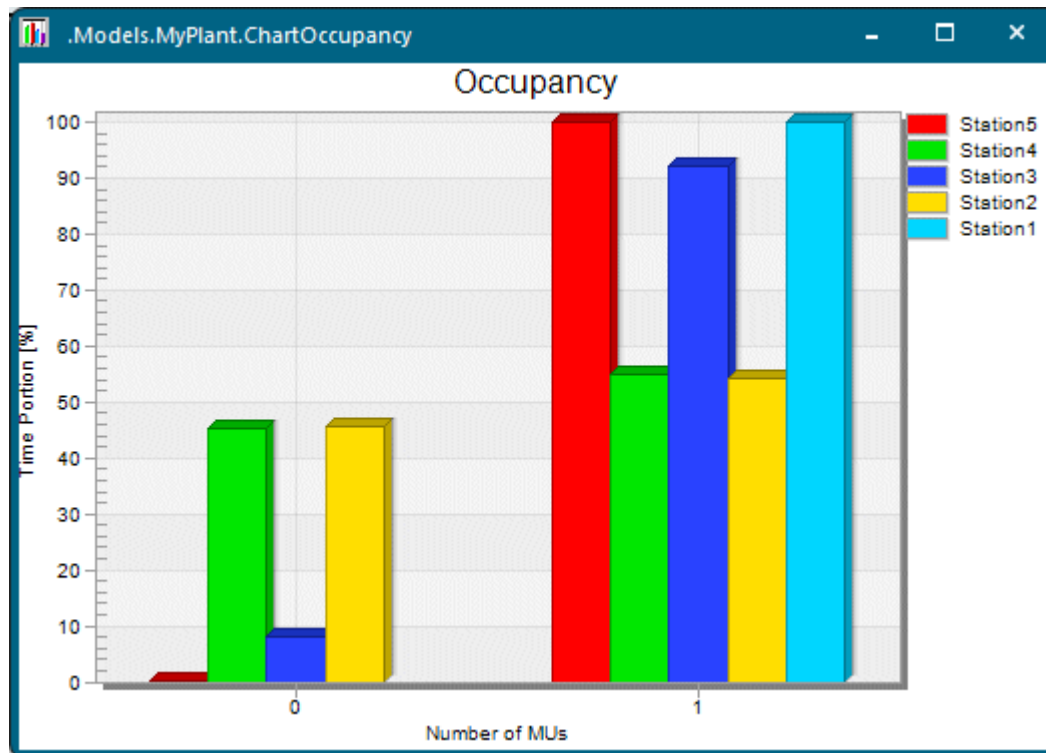
Remarks

You can select one of these settings:

- **None**
Adds no 3D effect to the **Chart Type** you selected.
- **Shadow**
Adds a black shadow to the chart type.

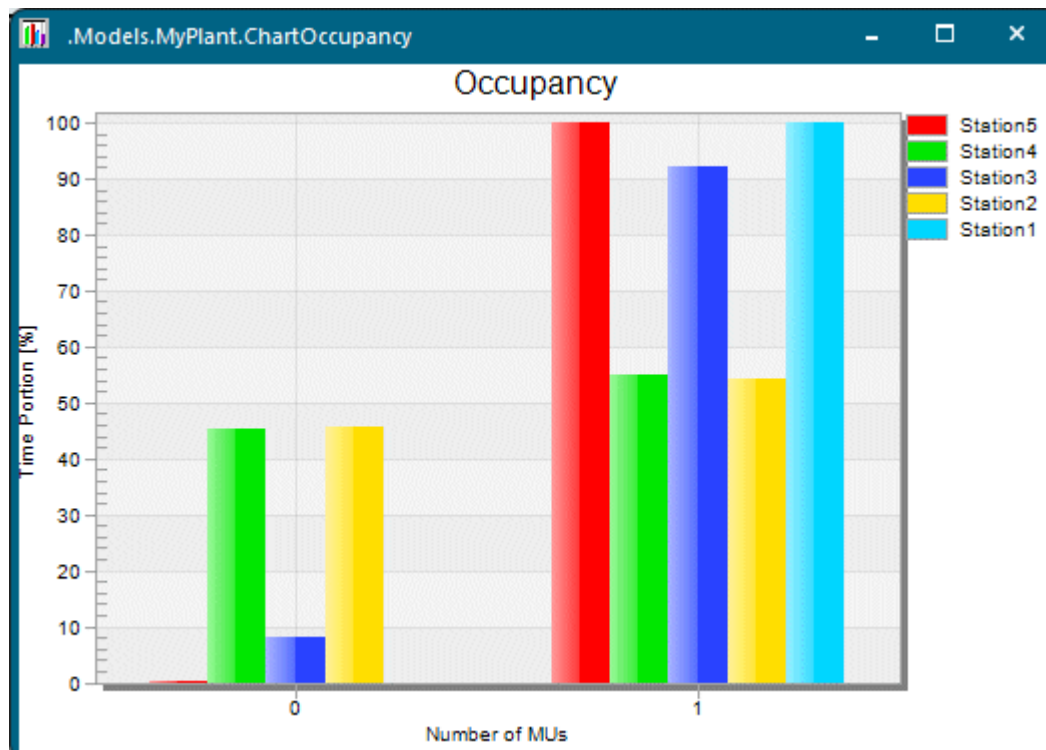


- **3D**
Adds three-dimensional depth to the chart type.



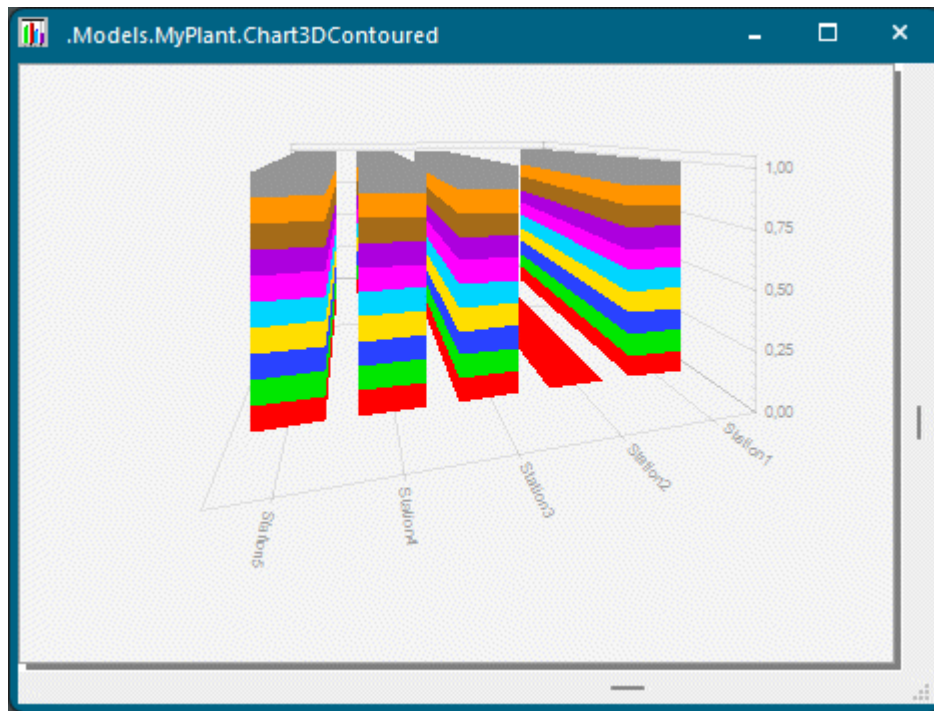
- **Gradient Bars**

Adds an effect between a gradient and a 3D look to the chart types **Columns** and **Bars**.



- **Contoured**

Applies to the chart types **3D Columns** and **3D Surface**.

**Note**

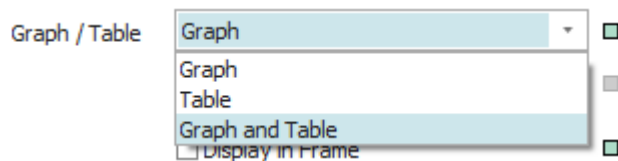
Some chart types do not provide **3D Effects**.

SimTalk

Effect [SimTalk]

Graph/table [drop-down list]

Select how the *Chart* shows the records.

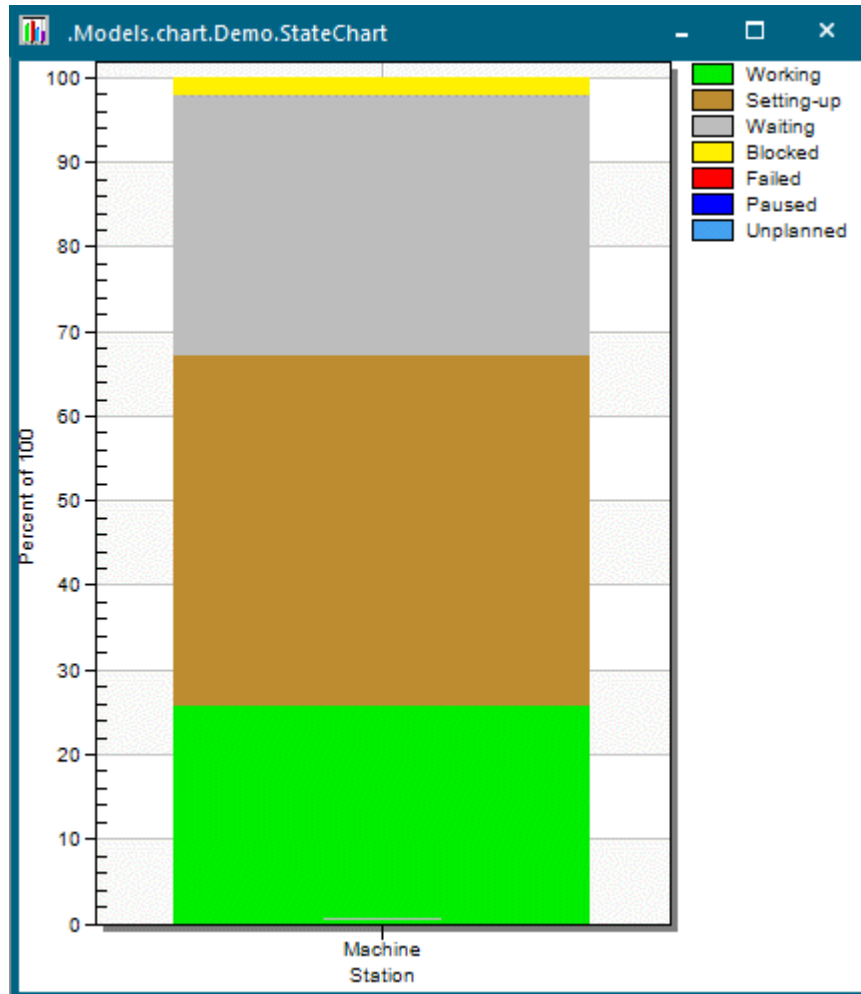


Remarks

You can select one of these settings:

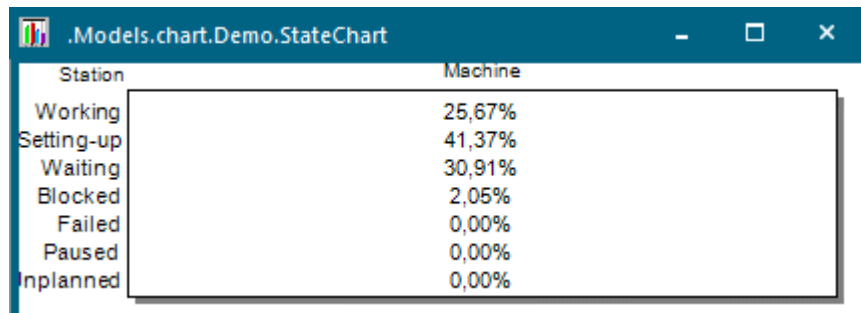
- **Graph**

Shows the records as a graph only in the *Chart* window.



- **Table**

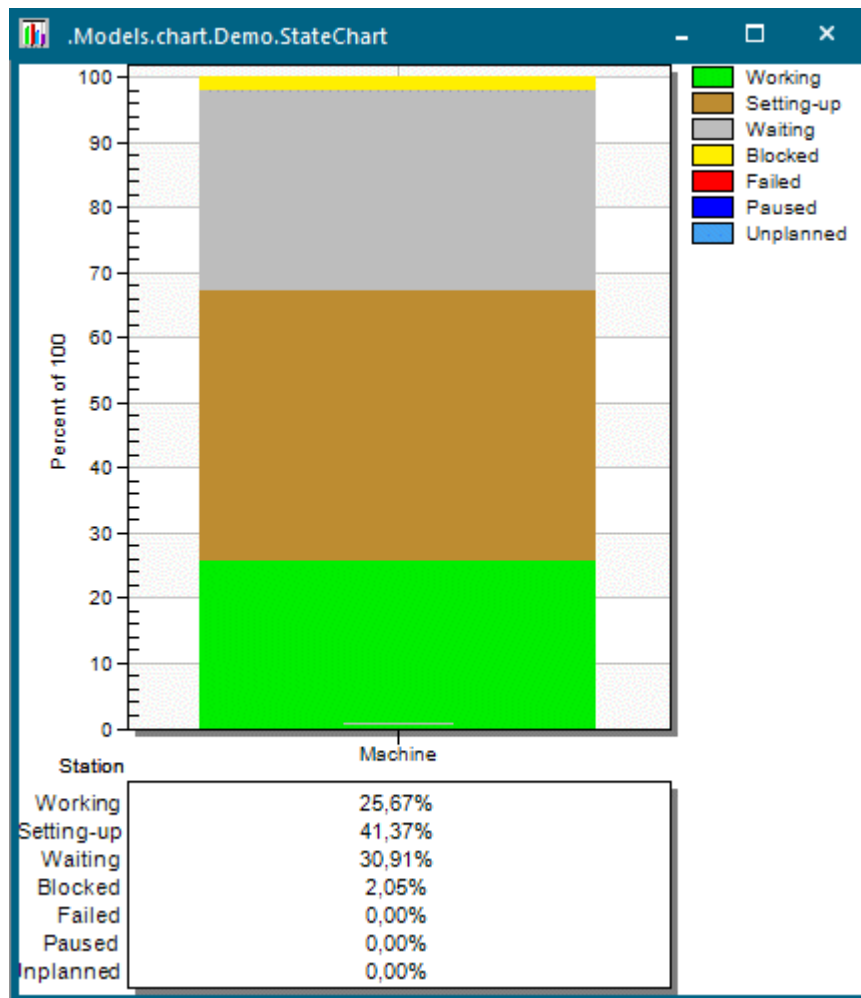
Shows the records as a table only in the *Chart* window.



Station	Machine
Working	25,67%
Setting-up	41,37%
Waiting	30,91%
Blocked	2,05%
Failed	0,00%
Paused	0,00%
Unplanned	0,00%

- **Graph and Table**

Shows the records as a graph and as a table in the *Chart* window.



SimTalk

[GraphTable \[SimTalk\]](#)

Display in Frame

Display in Frame [check box] - Chart

To display the *Chart* itself in the *Frame*, instead of the icon of the object, select this check box.

☒ Display in Frame

Width

2

Height

2

m

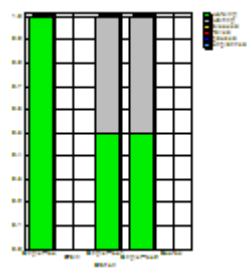
3D Image Quality

Balanced

Remarks

Then, type in the **Width** and the **Height** of the *Chart* and set the **3D Image Quality**.

We advise increasing the size of the icon of the *Chart* in the *Frame*: Hold down **Shift+Ctrl**, grab one of the four corners of the icon with the mouse and drag the icon to a convenient size.

Display in Frame On	Display in Frame Off
	

Note

When you select **Display in Frame**, you cannot freely combine **Line Styles** and **Line Weights** of the *Chart*.

SimTalk

DisplayInFrame [SimTalk]

Width [text box] - display in frame

Activate **Display in Frame** and type in the **Width** and the **Height** of the *Chart* in meters.

☒ Display in Frame 

Width Height  m

3D Image Quality 

SimTalk

[SizeInFrame \[SimTalk\]](#)

See also

[Display in Frame \[check box\] - Chart](#)

[Height \[text box\] - display in frame](#)

[Height \[text box\] - display in frame](#)

Height [text box] - display in frame

Activate **Display in Frame** and type in the **Height** and the **Width** of the *Chart* in meters.

☒ Display in Frame 

Width Height  m

3D Image Quality 

SimTalk

[SizeInFrame \[SimTalk\]](#)

See also

[Display in Frame \[check box\] - Chart](#)

[Width \[text box\] - display in frame](#)

3D Image Quality [drop-down list]

Select the **3D Image Quality** with which the *Chart* shows the image in 3D.

☒ Display in Frame

Width Height ☐ m

3D Image Quality

Balanced
High quality
Balanced
High performance

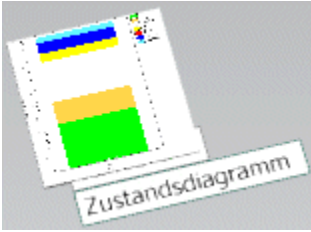
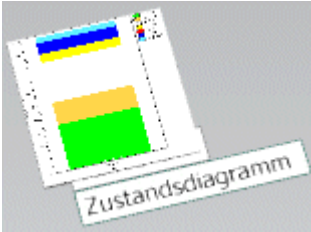
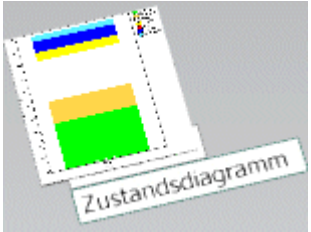
☐

Remarks

3D Image Quality is only available when you activate **Display in Frame**.

You can select one of these settings:

- **High quality**
Results in high resolution and improved image quality with bad performance.
- **Balanced**
Results in medium resolution, is a good compromise between image quality and performance.
- **High performance**
Results in low resolution and bad image quality with better performance.

High quality	Balanced	High performance
		

SimTalk

[ImageQuality3D \[SimTalk\]](#)

[DisplayInFrame \[SimTalk\]](#)

See also

[Display in Frame \[check box\] - Chart](#)

Gap When Null [check box]

To interrupt the line, which the *Chart* draws for values that are not defined, select this check box. To continue drawing the line, clear the check box.

☒ Gap When Null ☐

Undefined Value

Type in the value that the *Chart* takes to be as an undefined value. If a point takes this value, the *Chart* does not display it. The default value is **-999999**, i.e., a value that most likely none of the values to be displayed ever takes.



SimTalk

[GapWhenNull \[SimTalk\]](#)

[NullValue \[SimTalk\]](#)

Accumulated [check box] - Chart

To create an accumulated histogram, select this check box.

Category ☒ ☐ Accumulated

Remarks

The *Chart* then adds the sum of all previous values to each value. The last value thus will be 100 percent.

Note

The *Chart* only shows the check box for the **Category** > **Histogram**.

SimTalk

Accumulated [SimTalk]

Step Curve [check box]

To show a **Step Curve** instead of straight lines connecting the data points, select this check box.

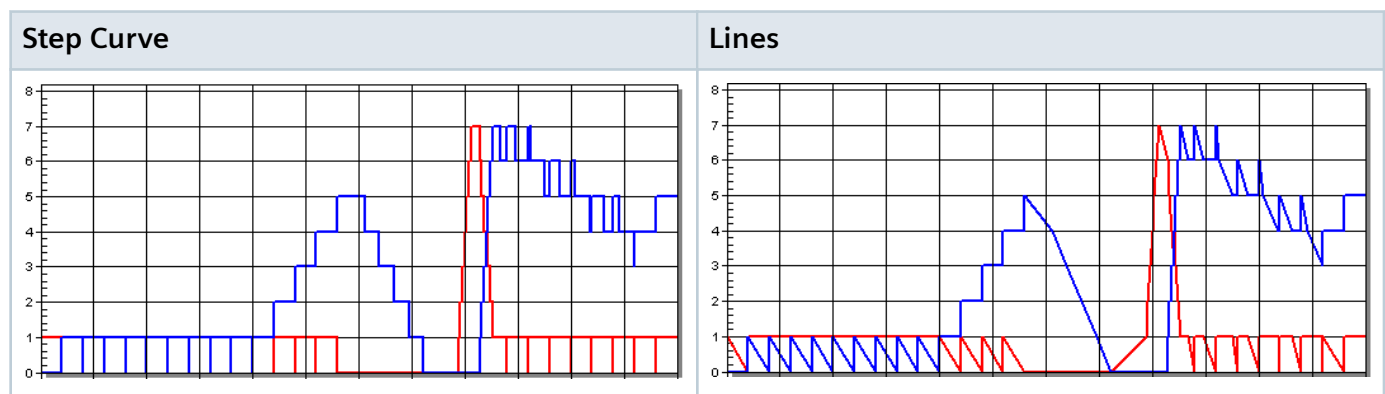
Category ☒ ☐ Step Curve

Remarks

A **Step Curve** might show some trends more clearly but might not quite be as accurate as the straight lines.

Note

The *Chart* only shows the check box for the **Category** > **Plotter**.



SimTalk

StepCurve [SimTalk]

Tab Axes

Tab Axes

Select how *Plant Simulation* shows the axes used in the *Chart* on the tab **Axes**.

The tab shows different settings depending on the **Category** you selected.

Data	Display	Axes	Labels	Color	Font	User-defined
Grid Lines						
		☒ Y-Axis			☐	Interval
		☒ X-Axis			☐	Number 25
Logarithmic						
		☐ Y-Axis			☐	
Range						
Y		0	...	∞	☐	
X		∞	...	∞	☐	

Data	Display	Axes	Labels	Color	Font	User-defined
Grid Lines						
		☒ Y-Axis			☐	Interval
		☒ X-Axis			☐	
Logarithmic						
		☐ Y-Axis			☐	
Number of Values						
		1000		☐ Scrollbar	☐	
Range						
Y		0	...	∞	☐	
X		1:00:00		Feed Rate (Grid Units)	6	☐

Grid Lines Y-Axis

To show **Grid Lines** on the **Y-Axis** in the display window of the *Chart*, select this check box. To hide them, clear the check box.

Grid Lines	☒ Y-Axis	☐	Interval	2
	☒ X-Axis	☐	Number	25

SimTalk

XGrid [SimTalk]

YGrid [SimTalk]

Interval

Type the **Interval** between the grid lines on the y-axis into the text box. It sets the distance of the horizontal grid lines and the labels on the y-axis.

Grid Lines	☒ Y-Axis	☐	Interval	2
	☒ X-Axis	☐	Number	25

If you do not type a value into the text box, *Plant Simulation* determines a meaningful interval based on the displayed values.

SimTalk

YGridLineInterval [SimTalk]

Grid Lines X-Axis

To show **Grid Lines** on the **X-Axis** in the display window of the *Chart*, select this check box. To hide them, clear the check box.

Type in the maximum **Number** of grid lines that the *Chart* shows on the x-axis.

Grid Lines	☒ Y-Axis	☐	Interval	2
	☒ X-Axis	☐	Number	25

SimTalk

[YGrid \[SimTalk\]](#)

[XGrid \[SimTalk\]](#)

[XGridlines \[SimTalk\]](#)

Number

Type in the **Number** of grid lines which the *Chart* shows on the x-axis. The *Chart* also shows a label for each grid line on the x-axis. If you are displaying a large number of data sets, it makes sense to enter a smaller number so that the labels can be shown completely.

If the number of grid lines is greater than the number of data sets on the x-axis, the *Chart* only shows as many grid lines as there are data sets.

If the number of grid lines is less than the number of data sets, the *Chart* reduces the number of grid lines and labels.

Note

For the **Category** > **XY-Graph**, the *Chart* only evaluates the **Number**, if you also type a fixed **Range X**, i.e., if you did not type in an asterisk *.

Grid Lines	☒ Y-Axis	☐	Interval	2
	☒ X-Axis	☐	Number	25

SimTalk

[XGridlines \[SimTalk\]](#)

Logarithmic Y-Axis

To activate **Logarithmic scaling** on the y-axis, select this check box.

Logarithmic ☐ Y-Axis ☒

Remarks

Logarithmic Y-Axis is useful if the data to be displayed is scattered across several orders of magnitude. Logarithmic representation cannot show negative values.

Note

If you activate **Logarithmic Y-Axis**, you also have to type in a value greater than 0, for example 0.1, for the **Range Y**, so that *Plant Simulation* can display the scale logarithmically.

SimTalk

YLog [SimTalk]

Logarithmic X-Axis

To activate **Logarithmic scaling** on the x-axis, select this check box.

Logarithmic	☒ Y-Axis	
	☒ X-Axis	

Remarks

Logarithmic X-Axis is useful if the data to be displayed is scattered across several orders of magnitude. Logarithmic representation cannot show negative values.

Note

Logarithmic X-Axis is only available for the **Chart Type > XY-Graph**.

Note

Plant Simulation does not show the check box **Logarithmic X-Axis** for all **Chart Types**.

SimTalk

[XLog \[SimTalk\]](#)

[Number of Values \[text box\]](#)

Number of Values [text box]

Type in the **Number of Values**, which the *Chart* collects over time per channel.

Number of Values ☐ Scrollbar ☒

Remarks

The setting **Number of Values** applies to the **Category > Plotter**.

The more values the *Chart* collects, the more memory *Plant Simulation* uses. Once the *Chart* reaches the number you entered, it starts dropping the least recent values.

SimTalk

[NumValues \[SimTalk\]](#)

See also

[Category \[drop-down list\]](#)

Scrollbar [check box]

To add a scrollbar to the *Chart* window for the **Category > Plotter**, select this check box. Clear the check box to not add a scrollbar.

Number of Values ☐ Scrollbar ☒

Remarks

If the scrollbar is active, you can type in a **Feed Rate** in grid units. The **Feed Rate** sets the number of grid units by which the *Chart* scrolls, when it reaches the right border of the display.

SimTalk

ScrollBar [SimTalk]

See also

Category [drop-down list]

Feed Rate [text box]

Type in the **Feed Rate** in grid units. It sets the number of grid units by which the *Chart* scrolls, once it reaches the right border of the display.

Feed Rate (Grid Units) 

Remarks

The setting **Feed Rate** applies to the **Category > Plotter**.

You can type in a value between 0.1 and 12.

Type in 0.1 for smooth scrolling, type in 12 for scrolling the entire page.

SimTalk

FeedRate [SimTalk]

See also

[Category \[drop-down list\]](#)

Step Size [text box]

Type in a **Step Size**. This is the width of the intervals into which the *Chart* divides the frequency distribution.

Step Size

Remarks

The setting **Step Size** applies to the **Category > Histogram**.

SimTalk

[StepSize \[SimTalk\]](#)

See also

[Category \[drop-down list\]](#)

Range Y

Type in the first value of the **Range** the *Chart* is to display on the y-axis into the left text box.

Range

Y

0

...

102

X

0

...

*

Remarks

The default setting 0 . . . * means that *Plant Simulation* shows the vales between 0 and the largest displayed value.

Note

When you activate **Logarithmic Y-Axis**, you also have to type in a value greater than 0, for example 0 . 1, for the **Range Y**, so that *Plant Simulation* can display the scale logarithmically.

Type in the last value of the range the *Chart* is to display scaled on the y-axis into the right text box.

Use the asterisk * to designate open or semi-open ranges along the y-axis and along the x-axis.

Compare these examples:

Type in	To do this
* ... *	Show the values between the smallest and the greatest displayed value on the y-axis.
0 ... *	Show the values between 0 and the greatest displayed value on the y-axis.
* ... 7.5	Show the values between the smallest displayed value and 7.5 on the y-axis.
-20 ... 20	Show the values between -20 and +20 on the y-axis.

Note

If you specify neither **Title** nor **Subtitle**, the *Chart* might, under certain circumstances, cut off the upper part of the **Range Y**.

Type a blank space into the text box **Subtitle** to prevent this.

SimTalk

YScaleMin [SimTalk]

[YScaleMax \[SimTalk\]](#)

See also

[XScaleMin \[SimTalk\]](#)

[XScaleMax \[SimTalk\]](#)

Range X

Type in the first value of the **Range**, which the *Chart* is to display scaled on the x-axis in the left text box.

Range

Y

0

...

102

▢

X

0

...

*

▢

Remarks

Type the last value of the **Range**, which the *Chart* is to display scaled on the x-axis into the right text box. The default setting 0 ... 0 means that *Plant Simulation* shows the entire range of definition.

If you are only displaying part of the range, the *Chart* shows a horizontal scrollbar, allowing you to scroll through the entire range of definition.

For the **Category > Histogram**, the **Range X** directly defines the range of definition. Type 0 into the right text box, so that the *Chart* dynamically adjusts the upper bound. Type the size of the intervals, into which the *Chart* divides the frequency distribution, into the text box **Step size**.

For the **Category > Plotter**, the **Range X** defines the time span, which the *Chart* shows in the display. You can also define the **Feed Rate** by which the *Chart* scrolls when it reaches the right-hand side of the display. Type in a number between 0.1 and 12. The number 0.1 stands for smooth scrolling, while 12 scrolls the entire page.

Use the asterisk * to designate open or semi-open ranges along the x-axis. You can also use it to designate dynamic upper limits for the **Histogram**.

Compare these examples:

Type in	To do this
* ... *	Show the values between the smallest and the greatest displayed value.
0 ... *	Show the values between 0 and the greatest displayed value on the x-axis.
* ... 7.5	Show the values between the smallest displayed value and 7.5 on the x-axis.
-20 ... 20	Show the values between -20 and +20 on the x-axis.

Negative values and 0 are useful for charts of category **XY Graph**.

SimTalk

XScaleMin [SimTalk]

[XScaleMax \[SimTalk\]](#)

[XRange \[SimTalk\]](#)

See also

[FirstInterval \[SimTalk\]](#)

[NumIntervals \[SimTalk\]](#)

[StepSize \[SimTalk\]](#)

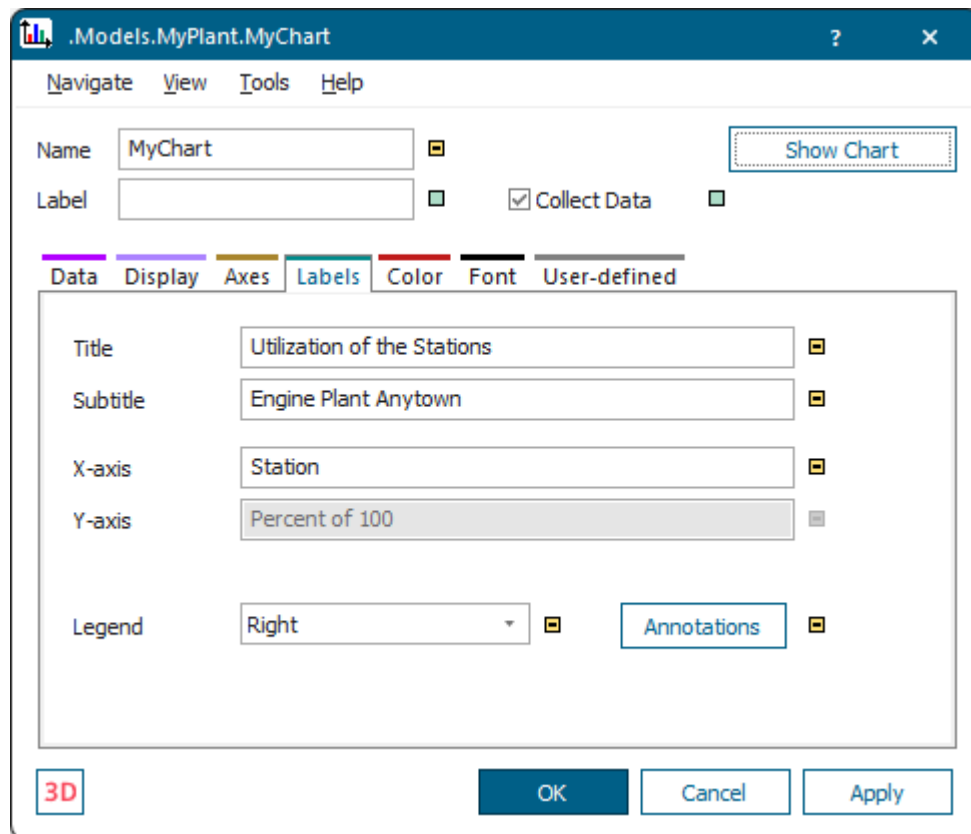
Tab Labels

Tab Labels [Chart]

Type in a **Title**, a **Subtitle**, and **labels** for the **X-Axis**, the **Y-Axis** and the **Z-Axis** of the *Chart*.

Remarks

Select if and where *Plant Simulation* shows the [Legend](#) and type in [Annotations](#) on the tab **Labels**.



Note

If you type in neither **Title** nor **Subtitle**, the *Chart* might, under certain circumstances, cut off the upper part of the **Y Range**. Enter a blank space into the text box **Subtitle** to prevent this.

See also

[Title \[text box\] - Chart](#)

[Subtitle \[text box\] - Chart](#)

[X-Axis \[text box\] - Chart](#)

[Y-Axis \[text box\] - Chart](#)

[Z-Axis \[text box\] - Chart](#)

Title [text box] - Chart

Type any text that *Plant Simulation* shows as the title of the *Chart* in the *Chart* window into the text box.

Title



Remarks

Select a font for the **Title**, select if you want to use **Bold** face and/or **Italic** face, and select a relative **Size** on the **Tab Font**.

SimTalk

Title [SimTalk]

Subtitle [text box] - Chart

Type any text that *Plant Simulation* shows as the subtitle of the *Chart* in the *Chart* window into the text box.

Subtitle

Engine Plant Anytown



Remarks

Select a font **Subtitle**, select if you want to use **Bold** face, and/or **Italic** face on the **Tab Font**.

SimTalk

SubTitle [SimTalk]

X-Axis [text box] - Chart

Type in any text that the *Chart* shows in the *Chart* window on the x-axis.

X-axis: 

SimTalk

[XLabel \[SimTalk\]](#)

Y-Axis [text box] - Chart

Type in any text that the *Chart* shows in the *Chart* window on the y-axis.

Y-axis: 

SimTalk

YLabel [SimTalk]

Z-Axis [text box] - Chart

Type in any text that *Chart* shows in the *Chart* window on the z-axis.

Z-axis:



Remarks

Note

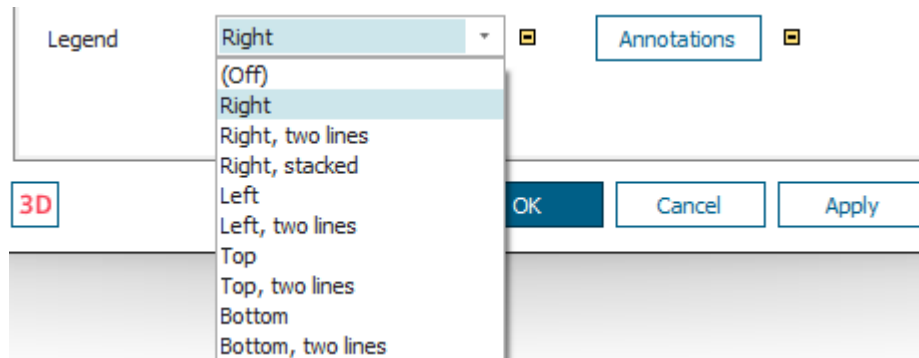
The *Chart* only shows the text box when you select one of the [3D Chart Types](#) > **3D Columns/3D Wire Frame** or **3D Surface**.

SimTalk

[ZLabel \[SimTalk\]](#)

Legend

Select if and on which side of the *Chart* window *Plant Simulation* shows the legend for the values which the *Chart* displays.



Remarks

You can select one of these settings:

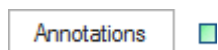
- **Off**, i.e., no legend.
- On the **Right**, in one line.
- On the **Right**, in **two lines**.
- On the **Right, stacked**.
Shows the input channels from bottom to top instead of from top to bottom. It is especially well suited for the [Chart Types](#) > **Stacked Columns** and **100 % Stacked Area**.
- On the **Left**, in one line.
- On the **Left**, in **two lines**.
- At the **Top**, in one line.
- At the **Top**, in **two lines**.
- At the **Bottom**, in one line.
- At the **Bottom**, in **two lines**.

SimTalk

[LegendLocation \[SimTalk\]](#)

Annotations [button]

To open the table **Annotations**, click this button. Then, add additional lines and text to the *Chart*.



Remarks

You can select these settings:

- **Type**

Select the **Type** of the annotation:

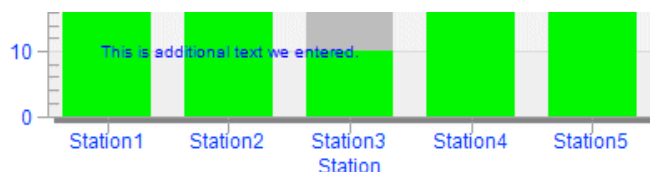
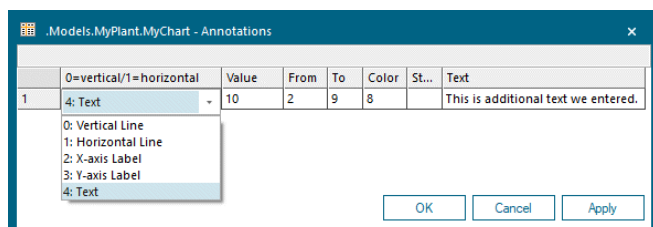
Select **0: Vertical Line** to show a vertical line.

Select **1: Horizontal Line** to show a horizontal line.

Select **2: X-axis Label** for labeling the x-axis.

Select **3: Y-axis Label** for labeling the y-axis.

Select **4: Text** for any text.



- **Value**

Type in the value where the *Chart* displays the line in the *Chart* window.

For a horizontal line or for an annotation of the Y-axis or for this is the Y-value.

For a vertical line or for an annotation of the X-axis this is X-value.

Type in Y-values in the Y scale of the *Chart*.

Type in X-values in the X scale of the *Chart*. For a bar chart the value 3.5 means that the X-value is placed between the third and the fourth bar.

- **From**

Type in the starting point of the line here. If you do not type in a value, the *Chart* extends an existing line.

This way you can not only create horizontal and vertical lines but also diagonal lines.

For a horizontal line or for text this is an X-value. For a vertical line this is a Y-value.

- **To**

Type in the end point of the line here.

If you do not type a value into the cells **From** and **To**, the *Chart* draws the line from the left to right, or from the top to the bottom of the display window.

For a horizontal line this is an X-value. For a vertical line this is a Y-value.

- **Color**

Type in the number of the color of the line here. Define colors on the [Tab Color](#).

- **Style**

For lines of **Type 0** or **1**, you can type in the line style.

For text of **Type 4**, you can type in the marker style.

Value	Line style	Marker style
0	a thin solid line	text only without a marker
1	a dashed line	a plus sign
2	a dotted line	a cross
3	a dash-dot line	a circle
4	a dash-dot-dot line	a filled circle
5	a medium thin solid line	a square
6	a thick solid line	a filled square
7	a grid tick	a diamond
8	a grid line	a filled diamond
9	none	upward triangle
10	a medium thick solid line	solid upward triangle
11	an extra thick solid line	downward triangle
12	none	solid downward triangle
13	none	small plus
14	none	small cross
15	none	small circle
16	none	small solid circle
17	none	small square
18	none	small solid square
19	none	small diamond
20	none	small solid diamond

Value	Line style	Marker style
21	none	small upward triangle
22	none	small solid upward triangle
23	none	small downward triangle
24	none	small solid downward triangle
25	none	large plus
26	none	large cross
27	none	large circle
28	none	large solid circle
29	none	large square
30	none	large solid square
31	none	large diamond
32	none	large solid diamond
33	none	large upward triangle
34	none	large solid upward triangle
35	none	large downward triangle
36	none	large solid downward triangle
92	none	arrow north
93	none	arrow north east
94	none	arrow east
95	none	arrow south east
96	none	arrow south
97	none	arrow south west
98	none	arrow west
99	none	arrow north west

- **Text**

Type in any text that describes the line here. You can optionally enter the following two-character prefix codes, consisting of the pipe symbol (|) and a letter. These control at which position in the graph the text will be positioned:

Prefix	Position
l	left inside edge of the graph
L	left outside edge of the graph
r	right inside edge of the graph
R	right outside edge of the graph
c	centered inside of the graph

Plant Simulation always shows user-defined annotations in the foreground. In previous versions this was not the case if the **Y-Axis** showed time values. As the *Chart* component, which *Plant Simulation* uses, cannot display times, *Plant Simulation* creates the captions for time values on the **Y-Axis** and the horizontal grid lines as **annotations**.

If you do add your own **annotations** to such a *Chart*, *Plant Simulation* now shows all **annotations**, including the grid lines, in the foreground. In this case, we recommend to hide the grid lines or to set the **Color** of the grid to a light color, for example to light gray.

SimTalk

[getAnnotations \[SimTalk\]](#)

[setAnnotations \[SimTalk\]](#)

See also

[Add Labels and a Legend](#)

Tab Color

Tab Color [Chart]

Select a **Color** for the **Grid**, the **Background**, the **Desk**, the **Text**, for the plotted data (numbers 1 through 14), and for additional colors on the tab **Color**.

Remarks

You can **Add** or **Delete** a color. You can also select the **Opacity**, a **Line Style**, and a **Line Weight**. For certain chart types you can also select a **Marker Size**.

If you would like the *Chart* to inherit color attributes, make sure that the **inheritance** check box for on the right-hand side of the tab looks like this ☒.

Data	Display	Axes	Labels	Color	Font	User-defined
+ Add ✗ Delete						
Color	Opacity	Line Style	Line Weight	Marker Type		
Grid	255	- ▾	- ▾	- ▾		
Background	255	-	-	-		
Desk	255	-	-	-		
Text	255	-	-	-		
1	255	▬ ▾	Medium ▾	Solid Circle ▾		
2	255	▬ ▾	Medium ▾	Solid Square ▾		
3	255	▬ ▾	Medium ▾	Solid Square ▾		
4	255	▬ ▾	Medium ▾	Solid Diamond ▾		
5	255	▬ ▾	Medium ▾	Diamond ▾		

SimTalk

[setColor \[SimTalk\]](#)

[setLineStyle \[SimTalk\]](#)

[getColor \[SimTalk\]](#)

[getLineStyle \[SimTalk\]](#)

See also

[Color \[Chart\]](#)

Color [Chart]

The *Chart* shows the number of the color or the name of item for which you define the color in the column **Color**.

Data	Display	Axes	Labels	Color	Font	User-defined
+ Add × Delete						
Color	Opacity	Line Style	Line Weight	Marker Type		
Grid	255	-	-	-		
Background	255	-	-	-		
Desk	255	-	-	-		
Text	255	-	-	-		
1	255	————	Medium	Solid Circle		
2	255	————	Medium	Solid Square		
3	255	————	Medium	Solid Square		
4	255	————	Medium	Solid Diamond		
5	255	————	Medium	Diamond		

Data	Display	Axes	Labels	Color	Font	User-defined
+ Add × Delete						
Color	Opacity	Line Style	Line Weight	Marker Type		
5	255	————	Thin	Plus		
6	255	————	Thin	Plus		
7	255	————	Thin	Plus		
8	255	————	Thin	Plus		
9	255	————	Thin	Plus		
10	255	————	Thin	Plus		
11	255	————	Thin	Plus		
12	255	————	Thin	Plus		
13	255	————	Thin	Plus		

Remarks

The tab shows these values:

- **Grid** (number -3)
Is the color the *Chart* uses for displaying the grid, i.e., the bounding rectangles, and the grid lines of the graph proper in the *Chart* window.
- **Background** (number -2)
Is the color the *Chart* uses for displaying the background of the graph proper in the *Chart* window.
- **Desk** (number -1)
Is the color the *Chart* uses for displaying the area located outside of the bounding rectangle of the graph's grid in the *Chart* window.
- **Text** (number 0)
Is the color the *Chart* uses for displaying text.
- **Colors** (numbers 1 through 14)
1 through 14 are the predefined colors of the input channels of the displayed data.

You can **add** Add... additional colors and define their properties to meet your modeling needs.

Double-click the **Color** box to open the MS Windows dialog **Colors** where you define colors.

You can select one of the predefined colors, or you can click **More Colors** and click the **Select** button to select a color in the color matrix. Then click **OK**. *Plant Simulation* shows this color next to **More Colors** and uses it as the active color.

SimTalk

setColor [SimTalk]

getColor [SimTalk]

Opacity

Type in a number for the opacity of the color. Type in 0 for an entirely transparent color, type in 255 for a completely opaque color.

Remarks

Semi-transparent colors will, for example, increase the readability of the **Chart Type > Area**.

Data	Display	Axes	Labels	Color	Font	User-defined
Add Delete						
Color	Opacity	Line Style	Line Weight	Marker Type		
Grid	255	-	-	-		
Background	255	-	-	-		
Desk	255	-	-	-		
Text	255	-	-	-		
1	255		Medium	Solid Circle		
2	255		Medium	Solid Square		
3	255		Medium	Solid Square		
4	255		Medium	Solid Diamond		
5	255		Medium	Diamond		

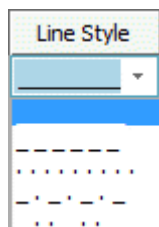
SimTalk

[setColor \[SimTalk\]](#)

[getColor \[SimTalk\]](#)

Line Style

To change the properties of the line with which the *Chart* shows the values, click the down-arrow in the cell whose property you want to change.



Remarks

After selecting the **Line Style**, you can select another **Line Weight** and another **Marker Type** from the drop-down lists.

Note

If you select **Display in Frame**, you cannot freely combine line styles and **Line Weights** of the *Chart*.

Select the line style with which the *Chart* shows the values.

The *Chart* uses these settings for the **Chart Types** > **Line** and **Spline**.

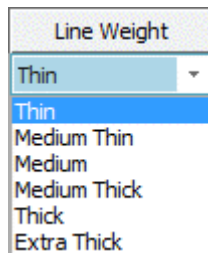
SimTalk

[setLineStyle \[SimTalk\]](#)

[getLineStyle \[SimTalk\]](#)

Line Weight

Select the **Line Weight** with which the *Chart* shows the values.



SimTalk

[setLineStyle \[SimTalk\]](#)

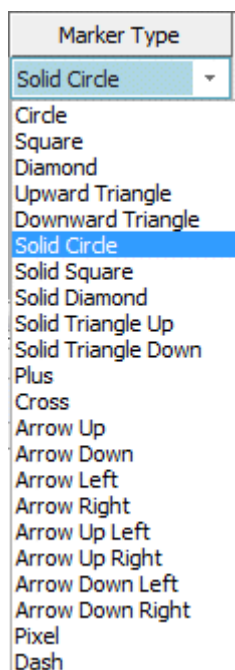
[getLineStyle \[SimTalk\]](#)

Marker Type

Select the **Marker Type** with which the *Chart* shows the values.

Remarks

The *Chart* uses these settings for the [Chart Types](#) > **Line with Markers** and **Spline with Markers**.



SimTalk

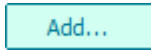
[setLineStyle \[SimTalk\]](#)

[getLineStyle \[SimTalk\]](#)

Add [color]

Add [color]

To add additional colors to the color table, click **Add**.



Remarks

Define a color in the dialog **Colors**, and click **OK** to add that color to the bottom of the **Color** table.

Double-click the **Color** box to open the MS Windows dialog **Color** where you define colors.

You can select one of the predefined colors, or you can click **More Colors** and click the **Select** button to select a color in the color matrix. Then click **OK**. *Plant Simulation* shows this color next to **More Colors** and uses it as the active color.

Delete [color]

To delete the color you selected from the **Color Table**, click this button.



See also

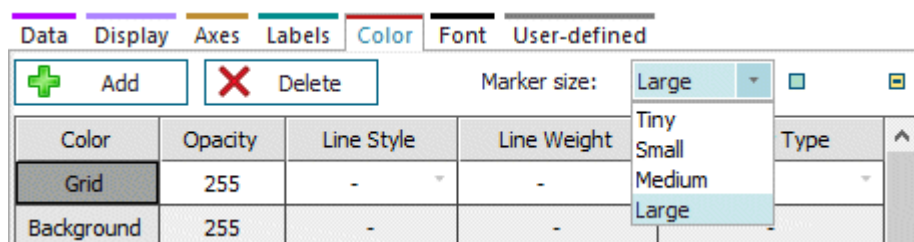
[Color \[Chart\]](#)

Marker Size

Select the size of the markers from the drop-down list.

Remarks

The setting **Marker Size** applies to the settings **Chart Types > Line with Markers** and **Spline with Markers**.



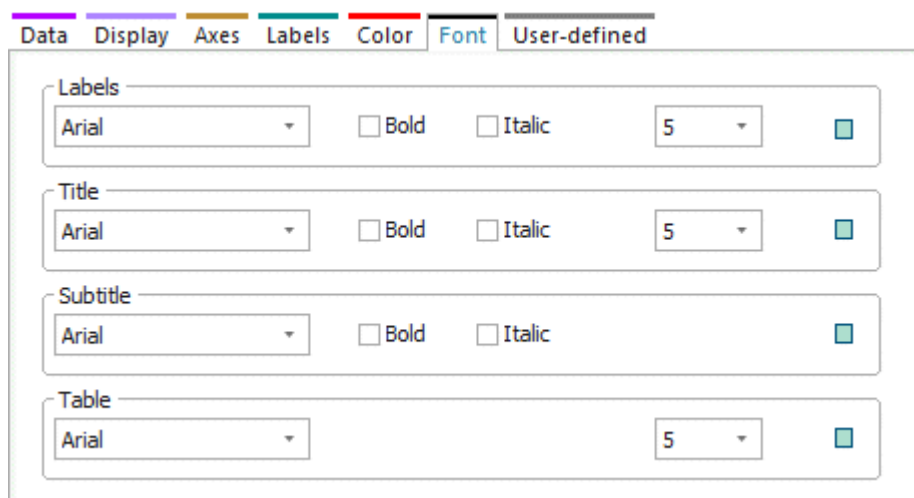
See also

[Chart Type \[drop-down list\]](#)

Tab Font

Tab Font [Chart]

Select a font, text effects and a zoom factor for the **Labels**, the **Title** and the **Subtitle**, and for data displayed as a **Table** on the tab **Font**.



Note

MS UI Gothic is intended for our **Japanese** users. The **English** or **German** version of Microsoft Windows might not provide this font by default.

See also

[Labels \[Chart\]](#)

[Title \[font\] - Chart](#)

[Subtitle \[font\] - Chart](#)

[Table \[font\] - Chart](#)

Labels [Chart]

Select a **font** from the drop-down list for the **labels**.

Labels

Arial ☐ Bold ☐ Italic 5 ☐

Remarks

This font does not apply to the **Title**, the **Subtitle** and to data displayed in **Table** format in the *Chart* window.

Bold

To apply **bold** face to all labels, select **Bold**. This does not apply to the **Title**, the **Subtitle** and to data displayed in **Table** format in the *Chart* window.

Italic

To *italicize* all labels, select **Italic**. This does not apply to the **Title**, the **Subtitle** and to data displayed in **Table** format in the *Chart* window.

Size

Select a size for the lettering for all labels. This does not apply to the **Title**, the **Subtitle** and to data displayed in **Table** format in the *Chart* window.

Note

The numbers in the drop-down list are not actual point sizes, but set a factor which the *Chart* uses for sizing labels.

SimTalk

LabelFont [SimTalk]

Title [font] - Chart

Select a font from the drop-down list for the **Title** of the *Chart* displayed in the *Chart* window.

The image shows a user interface for configuring the title font of a chart. It features a label 'Title' above a dropdown menu currently displaying 'Tahoma'. To the right of the font name are two checkboxes: 'Bold' and 'Italic', both of which are currently unchecked. Further right is a dropdown menu for font size, showing the number '5'. A small icon of a document with a pencil is located at the far right of the panel.

Bold

To apply **bold** face to the **Title** of the *Chart*, which is displayed in the *Chart* window, select **Bold**.

Italic

To *italicize* the **Title** of the *Chart*, which is displayed in the *Chart* window, select **Italic**.

Size

Select a size for the lettering from the drop-down list for the **Title** of the *Chart* displayed in the *Chart* window.

Note

The numbers in the drop-down list are not actual point sizes, but set a factor that the *Chart* uses for sizing labels.

SimTalk

TitleFont [SimTalk]

Subtitle [font] - Chart

Select a font from the drop-down list for the **Subtitle** of the *Chart*, which is displayed in the *Chart* window.



Bold

To apply **bold** face to the **Subtitle** of the *Chart*, which is displayed in the *Chart* window, select **Bold**.

Italic

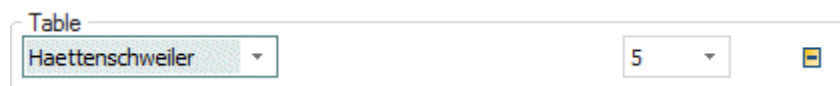
To *italicize* the **Subtitle** of the *Chart*, which is displayed in the *Chart* window, select **Italic**.

SimTalk

[SubtitleFont \[SimTalk\]](#)

Table [font] - Chart

Select a font from the drop-down list for data which the *Chart* shows in **Table** format in the *Chart* window.



Size

Select a size for data, which the *Chart* shows in **Table** format in the *Chart* window.

Note

The numbers in the drop-down list are not actual point sizes, but set a factor the *Chart* uses for sizing labels.

SimTalk

[TableFont \[SimTalk\]](#)

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

The commands are described under the [View Menu](#).

SimTalk

[updateDialog \[SimTalk\]](#)

Tools Menu

Tools Menu

The [Tools Menu](#) provides the commands listed in the table of contents to the left.

Use Metric Prefix [command]

Activates or deactivates if the *Chart* shows a unit prefix on the y-axis for very great or very small values, for example 100K instead of 100000.

The prefix	stands for
p	pico
n	nano
u	micro
m	milli
K	kilo
M	million
B	billion
T	trillion

SimTalk**UseMetricPrefix [SimTalk]**

Reset Values [command]

Deletes all values which the *Chart* collected.

SimTalk

[resetValues \[SimTalk\]](#)

Print [Chart]

Opens the dialog **Print**. Select a printer from the drop-down list.

Click **Setup** to open the **Properties** dialog of the selected printer and modify settings there.

SimTalk

[showPrintDialog \[SimTalk\] - Chart](#)

Copy to Clipboard

Copies the content of the *Chart* window as a bitmap to the clipboard.

SimTalk

`copyBitmapToClipboard [SimTalk] - Chart`

Help Menu

The commands are described under the [Help Menu](#).

Methods of the Chart

_Methods of the Chart

The *Chart* provides:

- The *methods* listed in the table of contents to the left.
- The [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			((Contents:table)) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				(IconName:string, Width:integer, Height:integer) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method Identifier and data type of the parameter you enter Data type of the return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			([Index:integer:=1]) -> object
MUPart	VOID			([Index:integer:=1]) -> object

Name of the method Identifier and data type of the parameter you enter Default value Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).